

Chapter 67

Working in the Node Editor

This chapter discusses how to work in the Node Editor, including multiple ways to add, connect, rearrange, and remove nodes to create any effect you can think of.

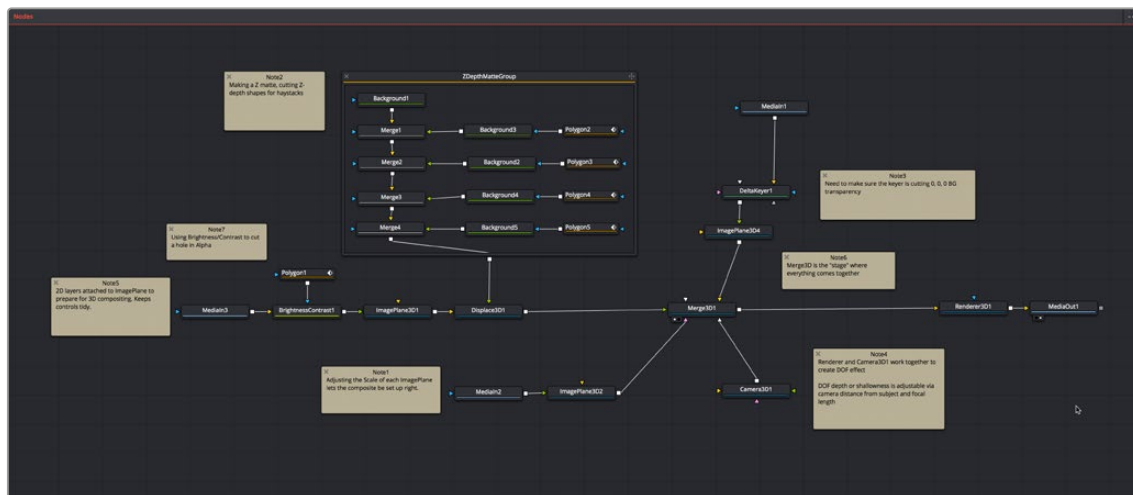
Contents

Learning to Use the Node Editor	1318
Navigating within the Node Editor	1319
Automatic Node Editor Navigation	1319
Using the Node Navigator	1319
Node View Bookmarks	1320
Adding Nodes to a Composition	1322
Adding, Inserting, and Replacing Nodes Using the Toolbar	1322
Adding Nodes Quickly Using the Select Tool Window	1323
Adding Nodes from the Effects Library	1324
Adding, Inserting, and Replacing Nodes Using the Contextual Menu	1327
Deleting Nodes	1328
Disconnected Nodes	1328
Selecting and Deselecting Nodes	1328
Loading Nodes into Viewers	1329
Viewed Nodes When You First Open Fusion	1330
Node View Indicators	1331
Drag and Drop Nodes into a Viewer	1331
Using the Contextual Menu	1331
Clearing Viewers	1332
Create/Play Preview	1332
Connecting and Disconnecting Nodes	1332
Node Basics	1332
How to Connect Nodes	1333
Dropping Connections on Top of Nodes	1333
Identifying Node Inputs	1335
Node Order Matters	1335

Upstream and Downstream Nodes	1338
Disconnecting and Reconnecting Nodes	1338
Tracing Connections Through the Node Tree	1339
Branching	1340
Connecting Merge Nodes	1341
Automatically Creating a Merge Node When Adding Nodes	1342
Automatically Creating a Merge Node by Connecting Two Outputs	1343
Connection Options and Routers	1344
Using Routers to Reshape and Branch Connections	1345
Swapping Node Inputs	1346
Extracting and Inserting Nodes	1347
Cut, Copy, and Paste Nodes	1348
Cut, Copy, and Paste in the Node Editor	1348
Pasting Node Settings	1349
Copying and Pasting Nodes to and from Any Text Editor	1349
Using Multilayer Nodes (Studio Version Only)	1350
Viewer	1350
Nodes and Inspector	1351
Input Layer Controls	1352
Instancing Nodes	1352
Using Instanced Nodes	1353
De-Instancing and Re-Instancing Specific Parameters	1354
Keeping Node Trees Organized	1354
Moving Nodes	1354
Renaming Nodes	1355
Changing Node Colors	1356
Using Sticky Notes	1356
Using Underlay Boxes	1357
Node Thumbnails	1358
Choosing Which Nodes Show Thumbnails	1359
Switching Thumbnails between Images and Icons	1360
Finding Nodes	1360
Performing Simple Searches	1361
Using Regular Expressions	1361
Custom Node Settings	1362
Managing Saved Settings	1362
Resetting Defaults	1363
Saving and Loading Alternate Node Settings	1363
Adding Saved Settings from the File System	1363
Node Modes Including Disable and Lock	1364
Node Editor Options	1364
Node Tooltips and the Status Bar	1365

Learning to Use the Node Editor

The Node Editor (formerly called the Flow or Flow Editor) is the heart of Fusion's compositing interface. It uses a flowchart structure called a node tree that lets you build a composition out of interconnected nodes, as opposed to using layers in a layer list. Each clip you add to the composition, and each image-processing operation you apply to those clips, is added as a node, all of which are joined together with connections that propagate image data from one node to the next. Each individual node performs a relatively simple operation, but collectively they combine to let you create wonderfully complex results.

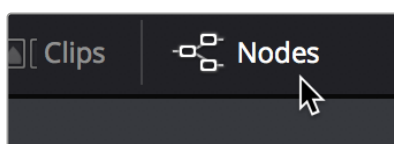


The Node Editor.

This chapter discusses how to work in the Node Editor in greater detail, showing you how to add, connect, rearrange, and remove nodes to create any effect you can think of.

To display the Node Editor:

- Click the Nodes button on the UI toolbar.



The Nodes button in the UI toolbar.

Navigating within the Node Editor

The Node Editor is the place where everything relating to nodes and the construction of your composites happens. The more you learn about how to navigate within the Node Editor, the faster you'll be able to work. There are a variety of standard methods of panning and zooming around the Node Editor, many of which are shared with other panels in Fusion.

Methods of panning the Node Editor:

- Middle-click and drag to pan around the Node Editor.
- Hold down Shift and Command, and then click and drag within the Node Editor to pan.
- Drag with two fingers on a track pad to pan in the Node Editor

Methods of zooming the Node Editor:

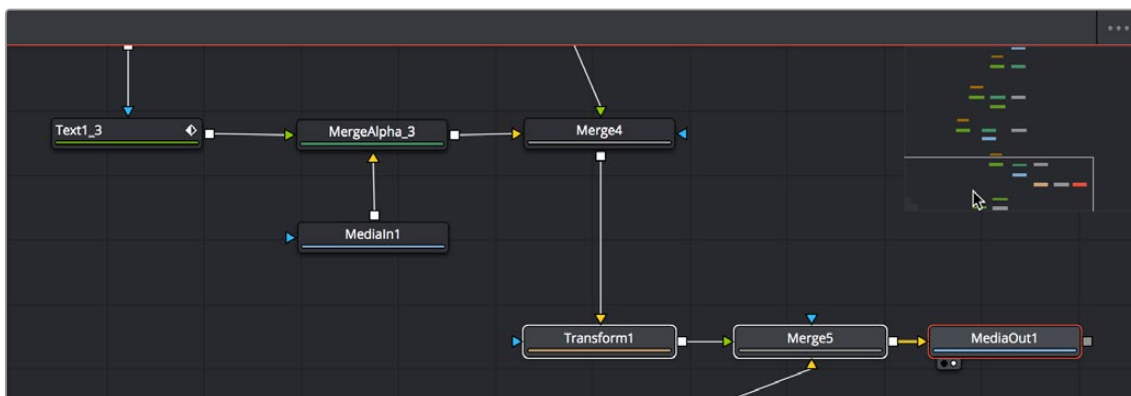
- Press the Middle and Left buttons simultaneously and drag to resize the Node Editor.
- Hold down the Command key and use your pointer's scroll control to resize the Node Editor.
- Right-click the Node Editor and choose an option from the Scale submenu of the contextual menu.
- Press Command-1 to reset the Node Editor to its default size.
- Hold down the Command key and drag with two fingers on a track pad to resize the Node Editor.

Automatic Node Editor Navigation

If a node that is not visible in the Node Editor becomes selected, either by using the Find command or by selecting a node's header in the Inspector, the Node Editor will automatically pan to display the node in the visible area.

Using the Node Navigator

Another useful way to pan around the Node Editor is to use the Node Navigator. The Node Navigator is a small rectangular overview in the upper-right corner of the Node Editor. It gives a bird's eye view of the entire composition, with an inner outline that indicates the portion of the composition that is visible in the panel. You can use the Node Navigator when you are zoomed in on a node tree and want to pan around a composition.



The Node Navigator.

To display or hide the Node Navigator, do one of the following:

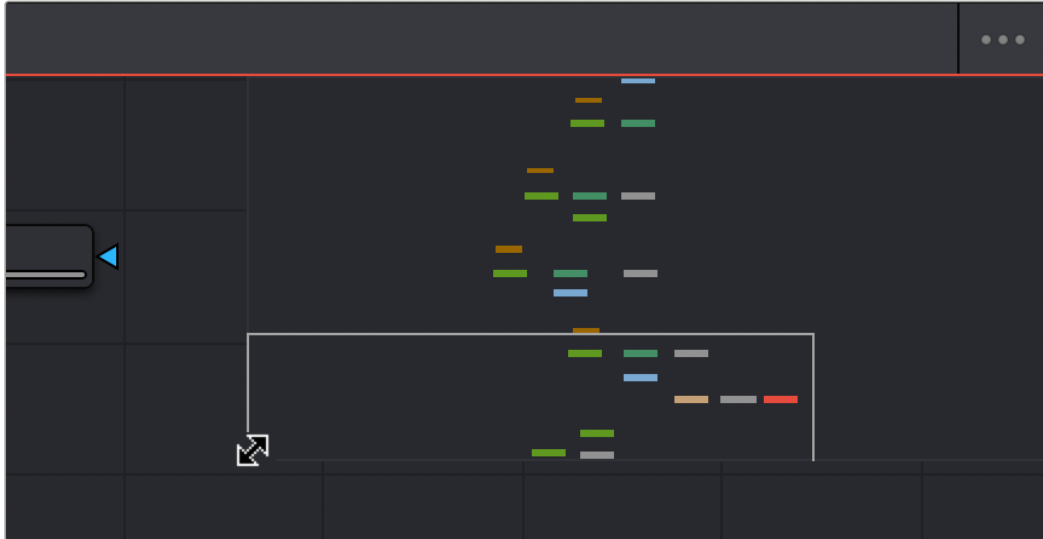
- Right-click in an empty area of the Node Editor, and then choose Options > Show Navigator.
- Press the V key.

To have the Node Navigator resume displaying automatically when needed after you've closed it:

- Right-click in an empty area of the Node Editor, and then choose Options > Auto Navigator.

To change the size of Node Navigator, do the following:

- Drag the lower-left corner of the Navigator to resize it.



Drag the corner to resize the Navigator.

To return to the default Node Navigator size, do the following:

- Right-click anywhere within the Node Navigator and choose Reset Size.

To pan the Node Editor using the Node Navigator, do the following:

- Drag within the Node Navigator to move around different parts of your node tree.
- Within the Navigator, drag with two fingers on a track pad to move around different parts of your node tree.

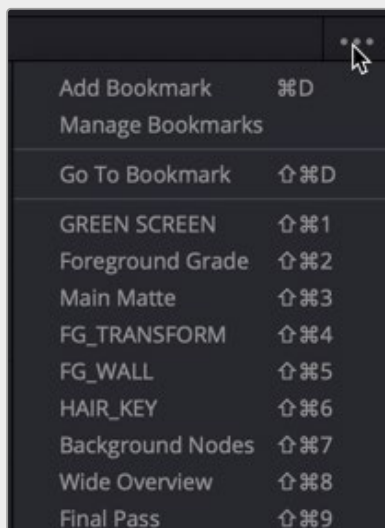
Node View Bookmarks

Bookmarks are another way of navigating the Node Editor. Bookmarks save the position and scale of the Node Editor, so you can quickly and precisely jump from viewing one group of nodes to viewing another.

To add a bookmark, do the following:

- 1 Pan and scale in the Node Editor to view a group of nodes you are interested in.
- 2 From the Options menu in the upper right of the Node Editor, choose Add Bookmark, or press Cmd-D.
- 3 In the Manage Bookmarks dialog that opens, enter a name for the bookmark and click the Add button.

The first nine saved bookmarks are given keyboard shortcuts and listed in the Options menu. They are also listed in the Go To Bookmarks dialog along with any saved bookmarks beyond the initial nine.



The Node Editor Options menu with nine bookmarks added

TIP: You can return the Node Editor to the default scale by right-clicking in the Node Editor and choosing Scale > Default Scale or pressing Cmd-1.

If your Node Tree changes and you want to update Bookmark names or delete bookmarks, those tasks can be done in the Manage Bookmarks dialog.

To rename or delete a bookmark, do the following:

- 1 From the Options menu in the upper right of the Node Editor, choose Manage Bookmarks.
- 2 In the Manage Bookmarks dialog that opens, right-click over the bookmark and choose Rename or Remove.
- 3 Click OK to close the Manage Bookmarks dialog.

Using Bookmarks

You can jump to a Bookmark view by selecting a bookmark listed in the Options menu or choosing Go To Bookmarks to open the Go To Bookmarks dialog. The Go To Bookmarks dialog has all the bookmarks listed in the order they were created in the current composition. Double-clicking on any entry in the dialog will update the Node Editor to that view and close the Go To Bookmarks dialog.

If you have a long list of bookmarks, you can use the search field at the bottom of the dialog to enter the name of the bookmark you want to find.

Changing the Sort Order and Assigning Keyboard Shortcuts

Bookmarks appear in the Options menu and in the Go To Bookmarks dialog in the order they were created. The top nine bookmarks listed are assigned keyboard short cuts. If you want to change the

sort order of the list you can do that in the Manage Bookmarks dialog. This can be useful if you want some bookmarks to have specific keyboard shortcuts, or you want a bookmark you made early in the process to not have a keyboard shortcut assigned to it.

To change the order of the listed bookmarks, do the following:

- 1 From the Options menu in the upper right of the Node Editor, choose Manage Bookmarks.
- 2 In the Manage Bookmarks dialog drag a bookmark up or down in the list.
- 3 An insert line appears where the bookmark will be inserted. Release the mouse button when the insert line is where you want the bookmark to be listed.

The keyboard short cuts will update to reflect the new order.

TIP: You can hold down the Shift key to select multiple bookmarks and move them simultaneously up or down in the Manage Bookmark list.

Using Underlays as Bookmarks

Underlays added to the Node Editor are automatically added as bookmarks. They are listed in the Options menu below the list of your custom bookmarks. You can omit Underlays from showing in the list of bookmarks by opening the Go To Bookmarks dialog and disabling the checkbox to Show Underlays. When the checkbox is disabled, Underlays will not show in the Go To Bookmarks dialog, and they will not appear in the Options menu.

Adding Nodes to a Composition

You can add nodes to the Node Editor in a variety of ways, depending on the type of node you're adding, and how much guidance you need to find what you're looking for. Additionally, the way you add nodes to a composition may also be dictated by how you need to attach that node to the current node tree.

Make Sure You're Adding Compatible Nodes

It's a good rule of thumb to make sure that whenever you're adding or inserting new nodes to the node tree, that you're adding nodes that are compatible with the nodes you're trying to attach to. For example, you'll have no problem inserting a Blur, Color, Filter, Paint, or Position node after almost any 2D operation. However, if you try to add a Merge3D node after a Glow node, it won't automatically connect, because those two nodes cannot be connected directly.

Adding, Inserting, and Replacing Nodes Using the Toolbar

The Fusion toolbar, located above the Node Editor, displays a selection of frequently-used nodes, displayed as buttons with distinct icons. These buttons make it fast to add Merge, Background, Paint, Mask, Transform, and many other commonly used nodes with the click of a button, or the drag of your pointer.



The Fusion page toolbar.

TIP: If you don't know which node a particular icon corresponds to, just hover your pointer over any toolbar button and a tooltip will display the full name of that tool.

Methods of adding nodes by clicking toolbar buttons:

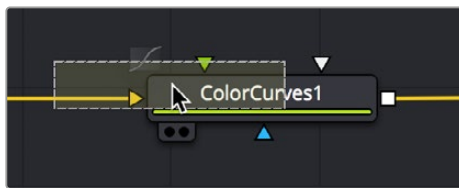
- **To add a node after a selected node:** Select a node in the Node Editor and then click a toolbar button.
- **To add a disconnected node to the Node Editor:** Deselect all nodes in the Node Editor and then click a toolbar button.

Methods of adding nodes by dragging toolbar buttons:

- **To insert a new node into the node tree:** Drag a toolbar button into the Node Editor and onto the connection line between any two compatible nodes. When the connection highlights as the node is over it, drop the node and it'll be inserted.
- **To create a disconnected node:** Drag a toolbar button into an empty part of the Node Editor. Dragging a toolbar button into the Inspector also creates a disconnected node.
- **To insert a new node after a node loaded into a viewer:** Drag a toolbar button onto a viewer to insert a new node after whichever node is viewed, regardless of whether any nodes are selected.

To replace a node in the Node Editor with a node from the toolbar:

- 1 Drag a button from the toolbar so that it's directly over the node in the Node Editor that you want replaced. When the node underneath is highlighted, drop the node.



Dragging a node from the toolbar to replace an existing tool.

- 2 Click OK in the dialog to confirm the replacement.

TIP: When you replace one node with another, any settings that are identical between the two nodes are copied into the new node. For example, replacing a Transform node with a Merge will copy the existing center and angle values from the Transform to the Merge.

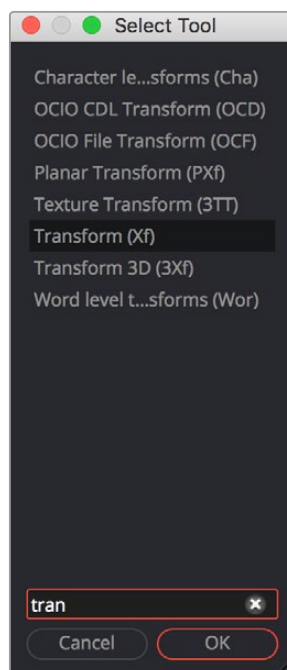
Adding Nodes Quickly Using the Select Tool Window

The next fastest way of adding or inserting nodes to the Node Editor is using the Select Tool window, which lets you search for any node available to Fusion by typing a few characters. Once you learn this method, it'll probably become one of your most frequently-used ways of adding nodes.

To use the Select Tool window to add nodes:

- 1 Do one of the following to determine if you want to insert a node or create a disconnected node:
 - If you want to insert a node, select a node that's compatible with the one you'll be creating, and the new node will be inserted after it.
 - If you want to create a disconnected node, deselect all nodes.

- 2 Press Shift-Spacebar to open the Select Tool dialog.
- 3 When the window appears, type characters corresponding to the name of the node you're looking for. A list automatically appears with likely candidates, and you can use the Up and Down arrow keys to select the correct node (if it's not selected already). If you don't know the name of the exact tool you're looking for, you can also type in common keywords and categories (i.e., keyer, materials, etc.) to get a selection of tools based on your input.
- 4 When you've selected the correct node, press the Return key (or click OK), and that node will be either inserted or added.



The Select Tool dialog lets you find any node quickly if you know its name.

TIP: Whenever you use the Select Tool window, the text you entered is remembered the next time you open it, so if you want to add another node of the same kind—for example, if you want to add two Blur nodes in a row—you can just press Shift-Spacebar and then press Return to add the second Blur node.

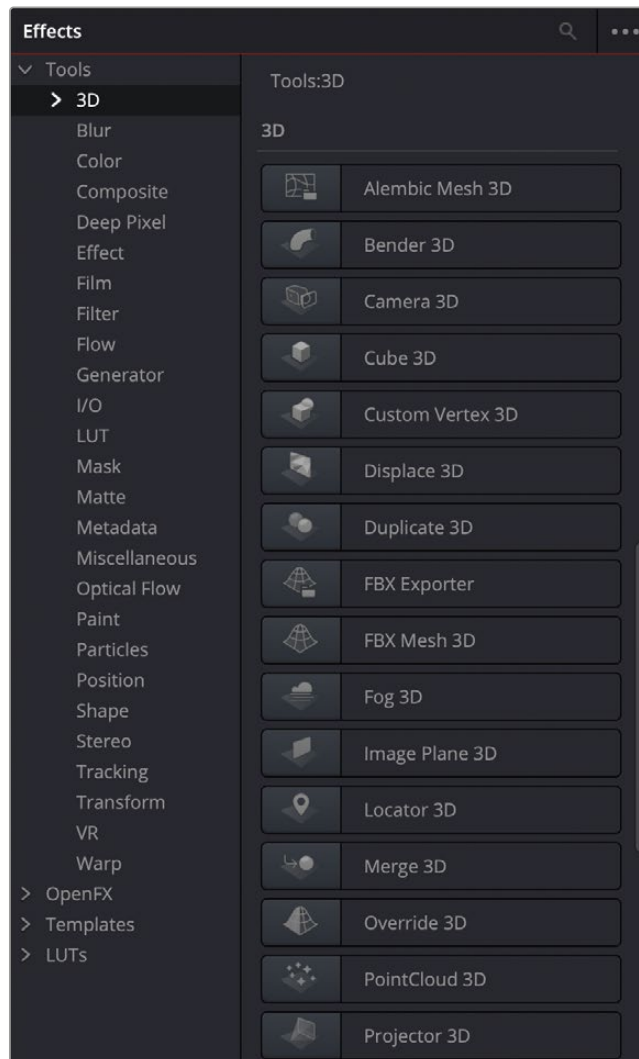
Adding Nodes from the Effects Library

While the toolbar shows many of the most common nodes you'll be using in any composition, the Effects Library contains every single tool available in Fusion, organized by category, with each node ready to be quickly added to the Node Editor. If you need more guidance to find the node you're looking for, or if you just want to browse around and see what's available, the Effects Library is the perfect place to start.

To open the Effects Library:

- Click the Effects Library button in the UI toolbar at the top of the Fusion window.

The Effects Library appears at the upper-left corner of the Fusion window, and consists of two panels. A category list at the left shows all categories of nodes and presets that are available, and a list at the right shows the full contents of each selected category.



The Tools bin of the Effects Library exposing 3D nodes.

By default, the category list shows the primary sets of effects: Tools, Open FX, Templates, and LUTs; with disclosure controls to the left that hierarchically show all subcategories within each category.

The categories are:

- **Tools:** Tools consist of all the effects nodes that you use to build compositions, organized by categories such as 3D, Blur, Filter, Mask, Particles, and so on.
- **Open FX:** All Resolve FX and any installed third party Open FX plugins will appear here.
- **Templates:** When using the Fusion page in DaVinci Resolve, templates consist of presets, macros, and utilities that have been created to get you started quickly. For example, Backgrounds consists of a variety of customizable generators that have been created using a combination of Fusion tools. Lens flares presents a wide variety of multi-element lens flares that you can add to any composition. Particles has a selection of pre-made particle systems that you can customize for your own use. Shaders has a variety of materials that you can use as texture maps for 3D text and geometry that you create in Fusion. And there are many, many other categories' worth of useful presets and macros that you can learn from and use in your own projects.
- **LUTs:** An assortment of pre-installed Look Up Tables for gamma, gamut, and colorspace conversions can be found here. Inserting a LUT into the Node Editor creates a new FileLUT (FLUT) node with the selected LUT file preloaded.

For more information on adding your own LUTs to this list, see *Chapter 150, “Using LUTs.”*

Adding, Inserting, and Replacing Tools Using the Effects Library

Adding nodes to the Node Editor from the Tools category of the Effects Library is very similar to adding nodes from the toolbar.

Methods of adding nodes by clicking in the Effects Library:

- **To add a node after a selected node:** Select a node in the Node Editor and then click a node in the browser of the Effects Library.
- **To add a disconnected node to the Node Editor:** Deselect all nodes in the Node Editor and then click a node in the browser of the Effects Library.

Methods of adding nodes by dragging from the Effects Library:

- **To insert a new node into the node tree:** Drag a node from the browser of the Effects Library into the Node Editor and onto the connection line between any two compatible nodes. When the connection highlights as the node is over it, drop the node and it'll be inserted.
- **To create a disconnected node:** Drag a node from the browser of the Effects Library into an empty part of the Node Editor. Dragging a toolbar button into the Inspector also creates a disconnected node.
- **To insert a new node after a node loaded into a viewer:** Drag a node from the browser of the Effects Library onto a viewer to insert a new node after whichever node is viewed, regardless of whether any nodes are selected.

To replace a node in the Node Editor with a node from the Effects Library:

- 1 Drag a node from the browser of the Effects Library so it's directly over the node in the Node Editor that you want replaced. When that node is highlighted, drop it.
- 2 Click OK in the dialog to confirm the replacement.

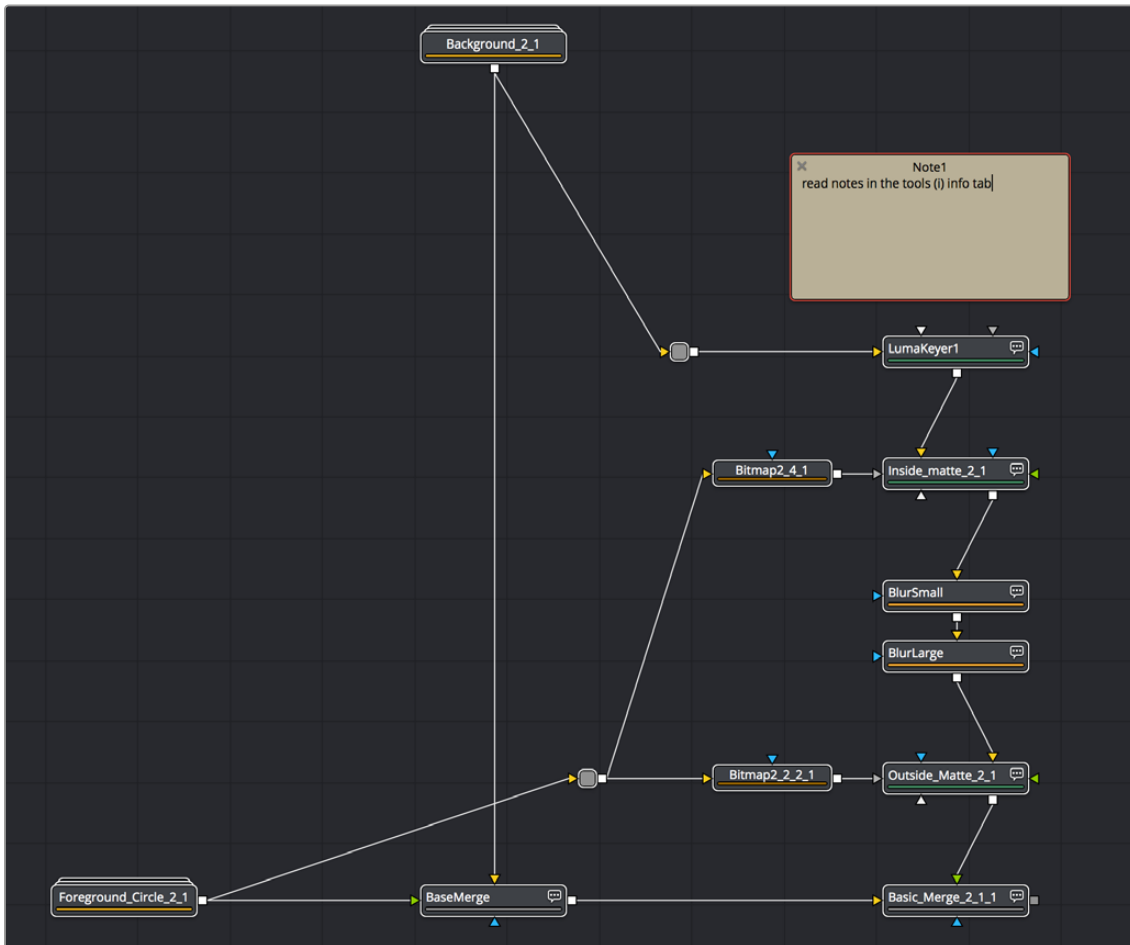
Adding, Inserting, and Replacing Templates Using the Effects Library

Adding items from DaVinci Resolve's Fusion page Templates category is often a bit different. Sometimes, as when adding a Lens Flare, a single node can be added or inserted into the Node Editor. When this is the case, adding nodes works the same as when adding from the Tools category.



Adding a Lens Flare effect.

Other times, such as when adding an item from the “How to” category, dragging a single item from the Node Editor results in a whole node tree being added to the Node Editor. Fortunately, all nodes of the incoming node tree are automatically selected when you do this, so it's easy to drag the entire node tree to another location in the Node Editor where there's more room. When this happens, the nodes of the incoming effect are exposed so you can reconnect and reconfigure it as necessary to integrate the effect with the rest of your composition.



Adding a LightWrap effect from the “How to” bin of the Templates category of the Effects Library.

Adding, Inserting, and Replacing Nodes Using the Contextual Menu

Another way of adding, inserting, and replacing nodes is to use the Node Editor’s contextual menu, which has dedicated submenus that let you create any kind of node available in Fusion. This can be a convenient when the pointer is already in the Node Editor selecting, moving, or connecting nodes.

Methods of adding nodes using the contextual menu:

- **To add a node:** Right-click in an empty area of the Node Editor, and choose a node from the Add Tool submenu.
- **To insert a node:** Right-click a node in the Node Editor, and choose a node from the Insert Tool submenu.
- **To replace a node:** Right-click a node in the Node Editor, and choose a node from the Replace Tool submenu.

TIP: When you replace one node with another, any settings that are identical between the two nodes are copied into the new node. For example, replacing a Transform node with a Merge will copy the existing center and angle values from the Transform to the Merge.

Deleting Nodes

To delete one or more selected nodes, press Delete (macOS) or Backspace (Windows), or right-click one or more selected nodes and choose Delete from the contextual menu. The node is removed from the Node Editor, and whichever nodes are connected to its primary input and output are now connected together. Nodes connected to other inputs (such as mask inputs) become disconnected.



Before deleting a node from a node tree (top), and after upstream and downstream nodes have automatically reconnected (bottom).

Disconnected Nodes

It's perfectly fine to have disconnected nodes, or even entire disconnected branches of a node tree, in the Node Editor alongside the rest of a composition. All disconnected nodes are simply ignored while being saved for possible future use. This can be useful when you're saving nodes that you've customized but later decided you don't need. It's also useful for saving branches of trees that you've since exported to be self-contained media that's re-imported to take the place of the original effect, but you want to save the original nodes just in case you need to redo your work.

Selecting and Deselecting Nodes

In order to work with nodes in the Node Editor in any way, or modify node parameters in the Inspector, you first need to learn to select the node or nodes you want to work with.

Selecting Nodes

Selecting nodes is one of the most fundamental things you can do to move nodes or target them for different operations. There are a variety of methods you can use.

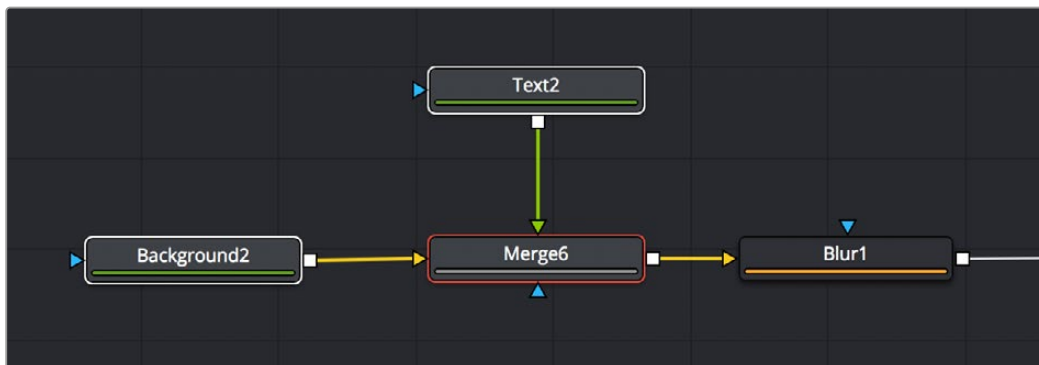
Methods of selecting nodes:

- **To select a single node:** Click any node in the Node Editor.
- **To select multiple nodes one at a time:** Command-click each node you want to select.
- **To select a whole region of nodes:** Drag a bounding box around all nodes you want to select.
- **To select all upstream or downstream nodes:** Right-click a node and choose Select > Upstream Nodes/Downstream Nodes from the contextual menu.
- **To select all nodes in the Node Editor:** Press Command-A.
- **To select a node from the Keyframe Editor:** Click any layer in the Keyframe Editor to select the corresponding node in the Node Editor.

The Active Node

When you select a single node using any of the methods described above, the selected node is known as the active node, and is highlighted orange to indicate that its parameters are currently editable in the Inspector (if the Inspector is open). This also indicates that node will be targeted for specific operations (such as inserting new nodes).

While multiple nodes can be selected, only one node will be the active node. To indicate the difference, the active node remains highlighted with orange, while all other selected nodes are highlighted with white. Unselected nodes have simple black outlines.



The active node is highlighted orange, while other selected nodes are highlighted white.

To set the active node when there are multiple selected nodes:

- Option-click one of the selected nodes in the Node Editor to make that one the active node.
- Open the Inspector (if necessary), and click a node's header bar to make it the active node.

Deselecting Nodes

Deselecting nodes, when necessary, works pretty much as you would expect.

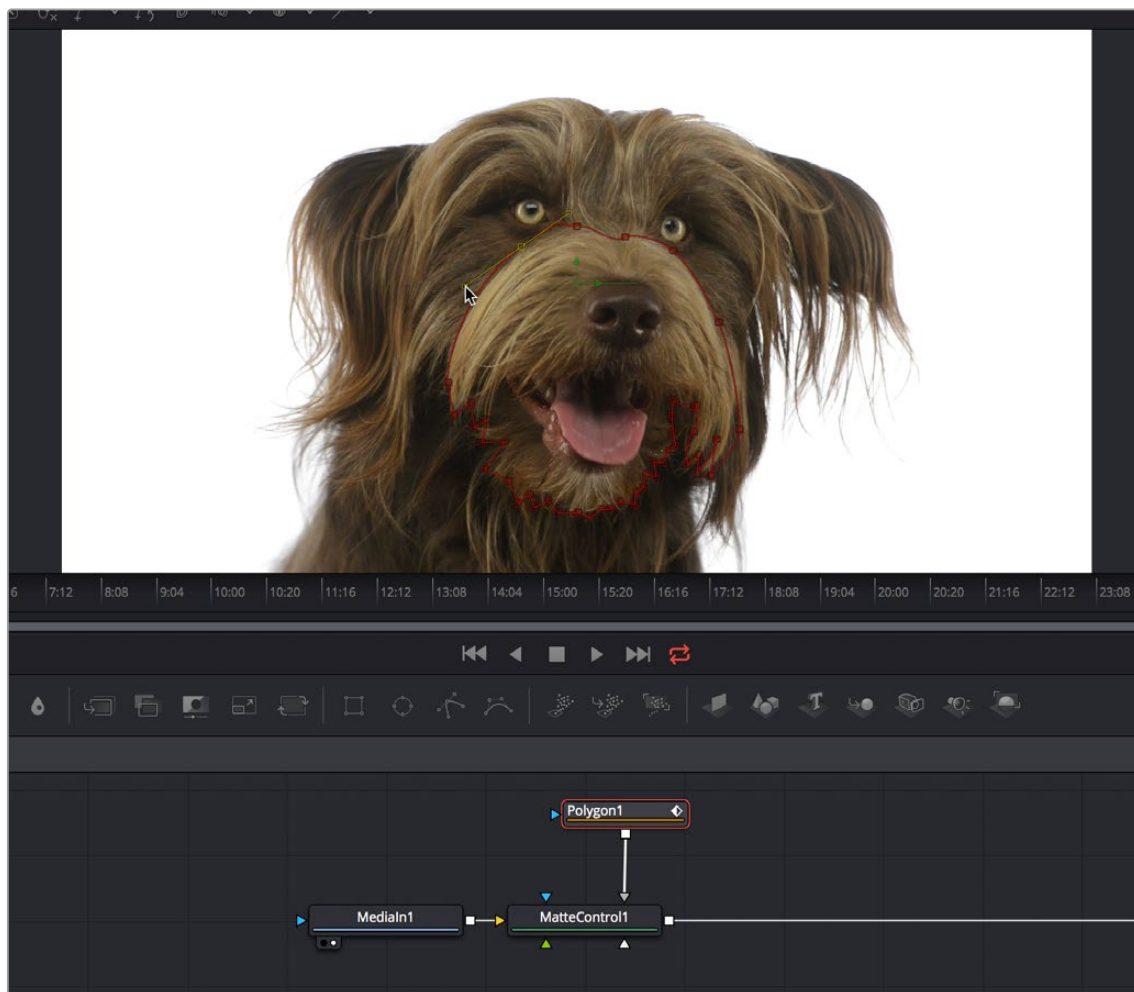
Methods of deselecting nodes:

- Click once in the background of the Node Editor to deselect all nodes.
- Press Command-Shift-A to deselect all nodes.
- Command-click to deselect multiple nodes one at a time.
- Command-drag a bounding box to deselect a group of selected nodes at one time.

Loading Nodes into Viewers

Once you've started building a composition, the next thing you need to learn is how to view specific nodes that you want to work on. This is important because the combination of which node is being viewed and which node is currently selected (these aren't always the same node) often determines which onscreen controls are available and how they appear.

In the following example, you're set up to rotoscope an image using a Polygon node that's attached to the garbage mask input of a MatteControl node which is inserting the mask as an alpha channel.



A node tree for doing a simple rotoscoping job.

As seen in the screenshot above, you'll want to load the upstream MediaIn or Loader node into a viewer while the Polygon node is selected for editing in order to see the full image you're rotoscoping while keeping the Polygon node's spline visible.

Viewed Nodes When You First Open Fusion

When you first open the Fusion page in DaVinci Resolve, the output of the current empty composition (the MediaOut1 node) is usually showing in viewer 2. If you're in Dual-viewer mode, viewer 1 remains empty until you assign a node to one of them.

When you first open Fusion Studio with an empty comp, both viewers remain empty even after reading in media using a Loader node. The viewers only display content when you assign a node to one of them.

There are several different ways to display a node in a viewer. Which ones you use depends on how you like to work.

Node View Indicators

The View indicators are displayed under each node, and serve two purposes. First, they're a clickable interface for displaying a node's output in one of the viewers. Second, they're an indication of which nodes in the Node Editor are being viewed. By default, there are two round indicators, representing the two viewers. The left and right indicators correspond to the left and right viewers, regardless of whether both viewers are visible or just one.



A viewer indicator enabled for the right viewer and disabled for the left viewer.

To load a node into a viewer using the Node View indicators:

- Clicking an indicator turns it white to show that node is currently loaded in the corresponding viewer. Clicking it again turns the indicator black and removes it from the viewer. Nodes only display View indicators if they're currently being viewed. If you want to view indicators, hovering the pointer over the node makes the indicators visible and available for clicking.
- You can also use keyboard shortcuts to toggle each View indicator. The default two viewers are assigned numeric keyboard shortcuts 1 and 2. Pressing the corresponding number once displays the selected node in the appropriate display view, while pressing it again clears that display.

For complex compositions, you may need to open additional viewers. For example, one viewer may be used to display the end result of the final comp, while another viewer displays the source, a third viewer displays a mask, and a fourth viewer might be a broadcast monitor connected via a Blackmagic DeckLink card or other display hardware. When you have more than two viewers, additional View indicators are added and each one is assigned a consecutive number between 3 and 9.

The more viewers you add, the more you may need help remembering which viewer is represented by which View indicator. Positioning the pointer over the View indicator in question will display a tooltip with the name of the viewer it represents.

Drag and Drop Nodes into a Viewer

If the View indicators are too small of a target for you to click on reliably and you are not keyboard oriented, another way to load a node into a viewer is to drag and drop it onto the viewer you want to load it into. This offers a quick explicit way to assign a node to a viewer, especially for pen and tablet users. Please note that as you drag, the node will appear to move at first, but it'll snap back into its original location once the pointer leaves the Node Editor.

Using the Contextual Menu

You can also right-click a node, and then choose View On > Left or Right to display the node on the appropriate viewer.

Clearing Viewers

Whenever you load a node into a viewer, you prompt that node, all upstream nodes, and other related nodes to be rendered. If you load nodes into both viewers, this is doubly true. If you want to prevent your computer from processing views that aren't currently necessary, you can clear each viewer.

Methods of clearing viewers:

- Press 1 or 2 to empty the left or right viewers if they're filled.
- Press ` (the Accent key) to empty both viewers.

Create/Play Preview

You can right-click a node, and choose an option from the Create/Preview Play On submenu of the contextual menu to render and play a preview of any node's output on one of the available viewers. The Render Settings dialog is displayed, and after accepting the settings, the tool will be rendered and the resulting frames stored in RAM for fast playback on that view.

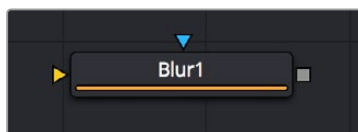
TIP: Hold the Shift key when selecting the viewer from the menu to bypass the Render dialog and to start creating the preview immediately using the default settings or the last settings used to create a preview.

Connecting and Disconnecting Nodes

Once you've started to add nodes to your composition, you need to connect them to perform their intended operations.

Node Basics

Each node displays small colored knots around the edges. One or more arrows represent inputs, and the square represent the tool's processed output, of which there is always only one. Outputs are white if they're connected properly, gray if they're disconnected, or red to let you know that something's wrong and the node cannot process properly.



A Blur node with a Foreground Input, Mask Input, and Output.

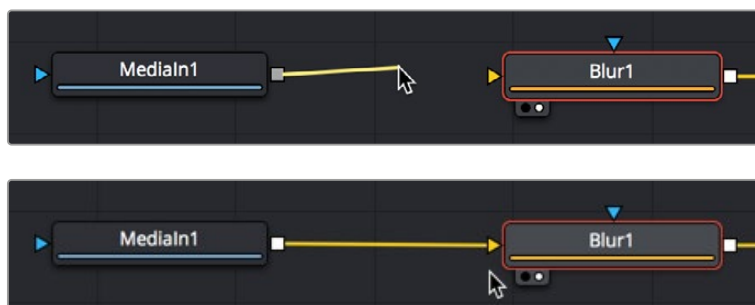
Each node takes as its input the output of the node before it. By connecting a MediaIn node's output to a Blur node, you move image data from the MediaIn node to the Blur node, which does something to process the image before the Blur node's output is in turn passed to the next node in the tree.



Two nodes connected together.

How to Connect Nodes

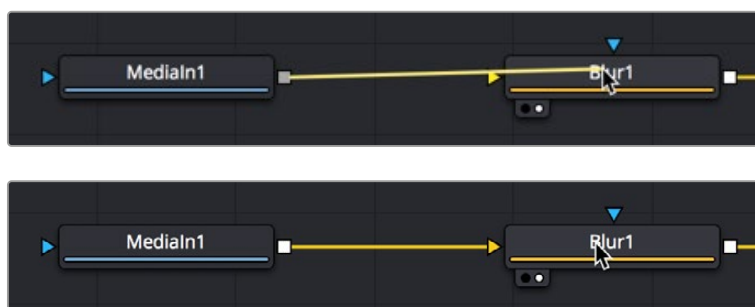
To manually connect one node to another, click on one node's output and drag a connection line out to drop on another node's input. The order in which you drag node connections is not important; you can just as easily drag a connection from one node's input to another node's output and get the same results.



Before (top), and after (bottom) dragging a connection line and dropping it to connect two nodes.

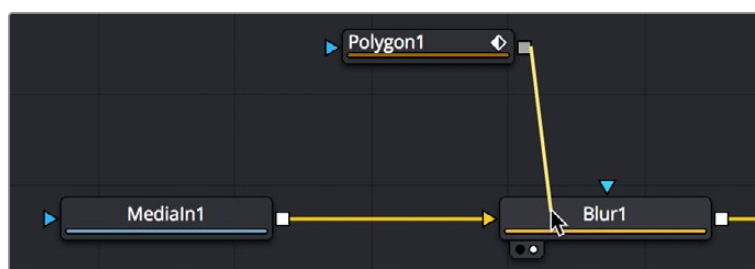
Dropping Connections on Top of Nodes

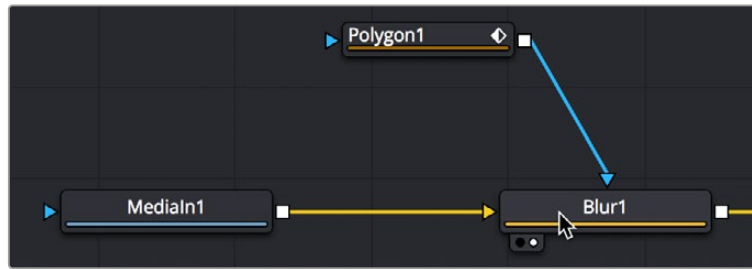
To make your life a bit easier, you can also drag a connection line and drop it directly on top of the tile of a node to automatically connect to the default input of that node, which is usually labeled “background” or “input.” In the following example, a connection is dragged from the output of a MediaIn node and dropped onto the tile of a Blur1 node, and the background input is connected first.



Before (top), and after (bottom) dragging a connection line and dropping it on top of a node.

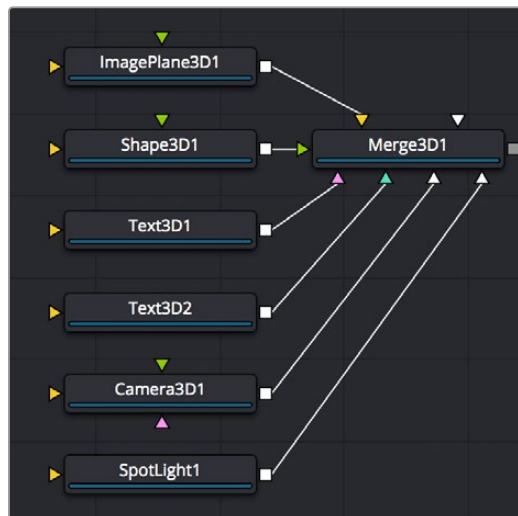
If you drop a connection on top of a node that already has the background input connected, then the second most important connection will be attached, which for multi-input nodes is the foreground input, and for other single-use nodes may be the Effects Mask input.





Before (previous page), and after (above) dragging a connection line and dropping it on top of a node that has the background input already connected.

Some multi-input nodes are capable of adding inputs to accommodate many connections, such as the Merge3D node. These nodes simply add another input whenever you drop a connection onto them.



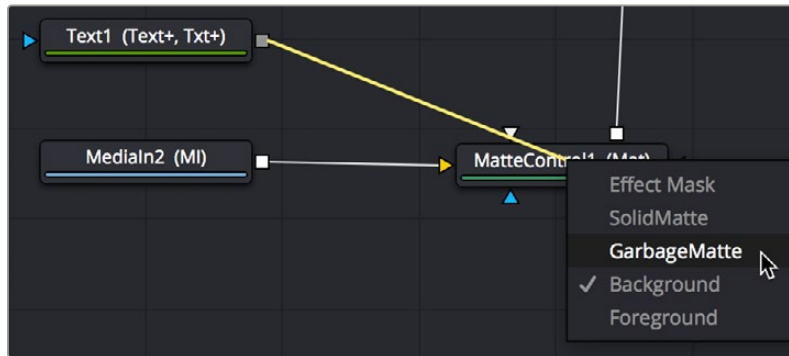
After dragging a connection line and dropping it on top of a Merge3D node.

Attaching Connections to Specific Inputs

If you want to make sure you don't attach a connection to the default input of a node, then you need to drop it right on top of the specific node input you want to attach it to. If you can see the input's label in the tooltip bar, then you know you're correctly positioned to make a good connection.

However, there's an alternate method of connecting nodes together in instances where there are several inputs to choose from and you want to make sure you're choosing the correct one. Hold down the Option key while dragging a connection from one node's output and dropping it onto the body of another node. This opens a pop-up menu from which you can choose the specific input you want to connect to, by name. Please note that this menu only appears after you've dropped the connection on the node and released your pointing device's button.





Option-dragging a node connection to drop onto another node exposes a node input menu.

Automatically and Manually Attaching Mask Nodes

Mask nodes, such as the Polygon, B-Spline, Ellipse, or Rectangle, have a different automatic behavior when you connect them to other nodes. If you drag a connection from a Mask node onto the body of another node, it will automatically connect itself to the default mask input, which is usually the effect mask input. The assumption is that you're using the mask to limit the node's effect somehow. However, this isn't always the case, so you'll need to be careful of this behavior to make sure you're attaching your mask to the input that will actually create the effect you need.



Before (left) and after (right) dragging a connection from a Mask node and dropping it on top of a MatteControl node.

Identifying Node Inputs

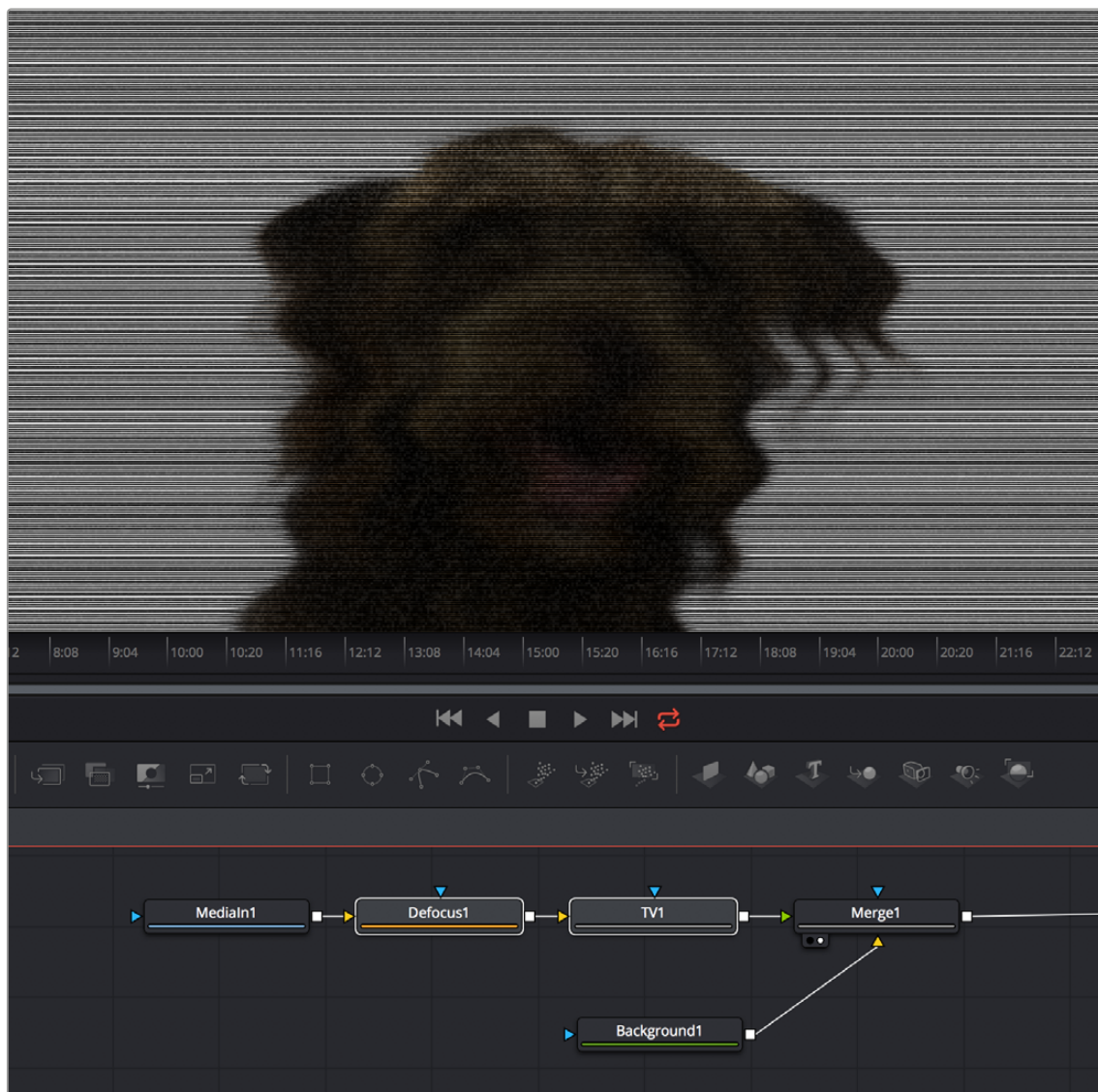
While you are still figuring out all the nodes and their inputs, hovering the pointer over any knot will display a node tip with the knot's name.

TIP: Rather than remembering the different knot types, press the right mouse button, hold Option, and drag from the output of a node to the center of another tool. When you release the mouse, a tooltip will appear allowing you to select the knot you want to connect to.

Node Order Matters

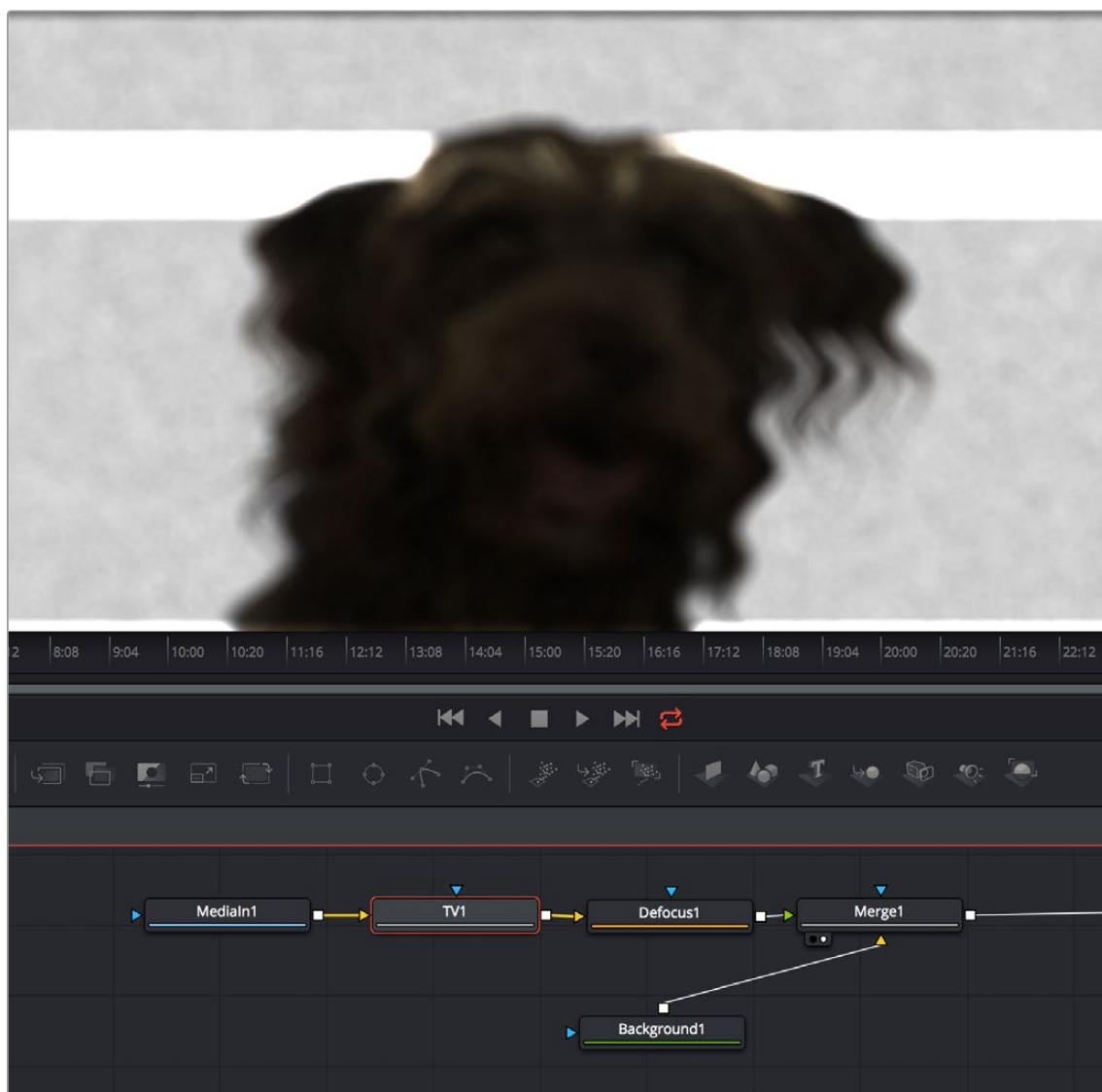
The order in which nodes are attached defines the order in which each image-processing operation is applied to the image.

In the following example, a MediaIn node adds a clip to the composition, while a Defocus node blurs the image, and then a TV node adds scanlines and vertical distortion. Those effect nodes are then connected to the MediaOut node in the Fusion page in DaVinci Resolve or a Saver node in Fusion Studio.



Adding a Defocus effect first, then the TV node second.

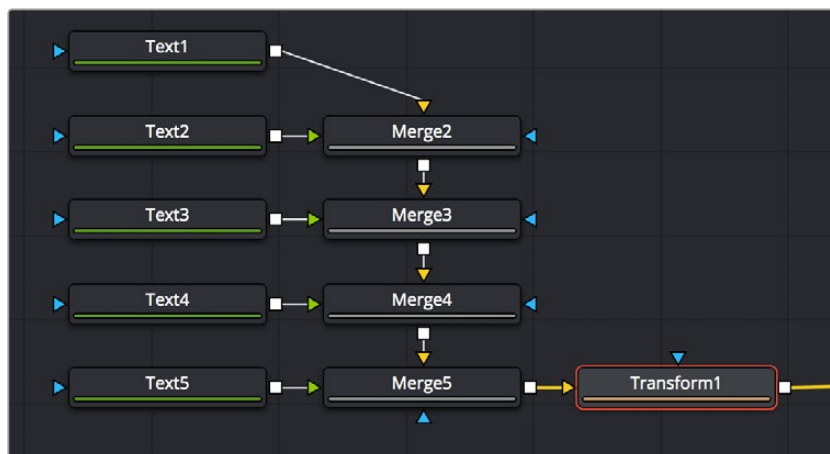
As you can see above, connecting the Defocus node first, followed by the TV node, means that while the initial image is softened, the TV effect is sharp. However, if you reverse the order of these two nodes, then the TV effect distorts the image, but the Defocus node now blurs the overall result, so that the TV effect is just as soft as the image it's applied to. The explicit order of operations you apply makes a big difference.



Adding a TV effect first, and a Defocus second.

As you can see, the node tree that comprises each composition is a schematic of operations with tremendous flexibility. Additionally, the node tree structure facilitates compositing by giving you the ability to direct each node's output into separate branches, which can be independently processed and later recombined in many different ways, to create increasingly complex composites while eliminating the need to precompose, nest, or otherwise compound layers together, which would impair the legibility of your composition.

In the following example, several graphics layers are individually transformed and combined with a series of Merge nodes. The result of the last Merge node is then transformed, allowing you to move the entire collection of previous layers around at once. Because each of these operations is clearly represented via the node tree, it's easy to see everything that's happening, and why.



The output of five Text nodes being combined using Merge nodes is modified by a single Transform node.

Upstream and Downstream Nodes

Since nodes can be positioned anywhere in the Node Editor, and added in any direction, nodes are referred to as being upstream and downstream of one another. Once you select a node, all other nodes that directly or indirectly connect to its input are considered to be upstream. Any other nodes that are directly or indirectly connected to the output are said to be downstream.

This is an important distinction to make because, unlike layer-based systems, the visual positioning of nodes in your node tree has no bearing on the order of operations in that composition. The only thing that matters is whether nodes are upstream or downstream of each other.



Tools upstream (left) and downstream (right) of the Merge node.

TIP: To help you stay organized, there are Select > Upstream/Downstream commands in the Node Editor contextual menu for selecting all upstream or downstream nodes to move them, group them, or perform other organizational tasks.

Disconnecting and Reconnecting Nodes

Node trees are a continuous work in progress, requiring constant revision and rearrangement as you discover new details that need to be finessed, or things that you can do better once the overall composition has taken shape. To facilitate quick changes, each connection between two nodes is divided into two halves: the output half (connected to the upstream node's output) and the input half (connected to the downstream node's input). This can only be seen when you hover the pointer over a connection. The half your pointer is over is highlighted in blue.



The two halves of a connection line that are revealed when you hover your pointer over it.

By clicking and/or dragging these two halves, it's possible to quickly disconnect, reconnect, and overwrite node connections, which is essential to rearranging your node tree quickly and efficiently.

To disconnect two nodes, do one of the following:

- Click once on the input half of the connection between two nodes.
- Click on the input arrow to which a connection is attached, then drag to pull the connection away from the tool and drop it anywhere in an empty area of the Node Editor.

To overwrite a current connection:

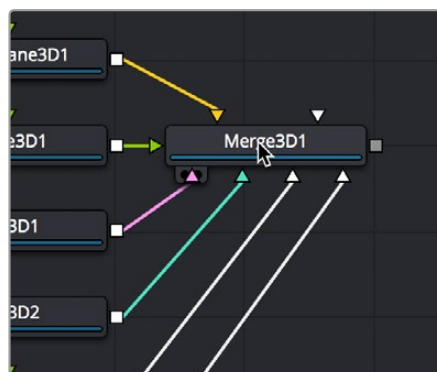
- Drag the output or input half of a connection, and drop it directly onto another node's input or output. This simultaneously disconnects the previous connection and connects the one you're dragging.

To reconnect a connection from one node to another:

- Drag the output or input half of a connection to disconnect it from one node, and drop the connection directly on another node's input or output.

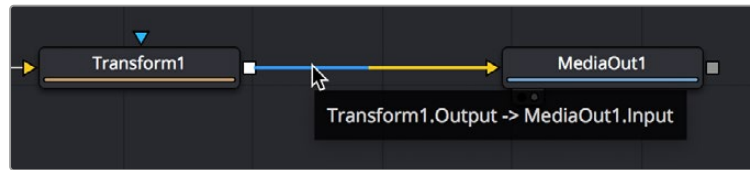
Tracing Connections Through the Node Tree

Positioning the pointer over a node causes the connections attached to that node to become highlighted, which makes it easier to see which nodes it's attached to. Additionally, highlighted connections display the color of the input they're connected to, which makes it easy to see if they're connected to a foreground, a background, or a particular kind of mask.



Hovering the pointer over a node highlights the color of all connections, telling you what kinds of inputs are connected.

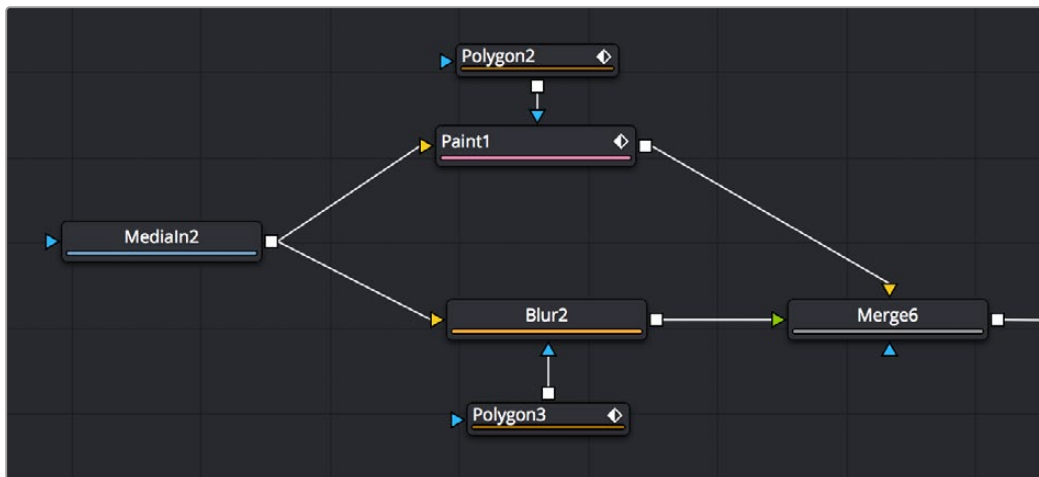
Additionally, positioning the pointer over a connection causes a tooltip to appear that displays the output and input that connection is attached to.



Hovering the pointer over a node highlights the connection between it and other nodes.

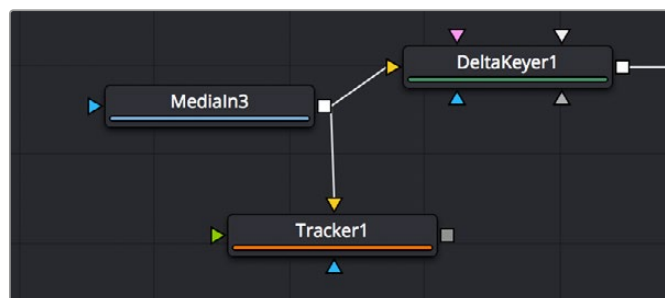
Branching

A node's input can only have one connection attached to it. However, a tool's output can be connected to inputs on as many nodes as you require. Splitting a node's output to inputs on multiple nodes is called branching. There are innumerable reasons why you might want to branch a node's output. A simple example is to process an image in several different ways before recombining these results later on in the node tree.



A MediaIn node branched to two node operations and then recombined using a Merge node.

Alternatively, it lets you use one image in several different ways—for example, feeding the RGB to one branch for keying and compositing, while feeding the A channel to the Effects Mask input of another node to limit its effect, or feeding RGB to a tracker to extract motion information.



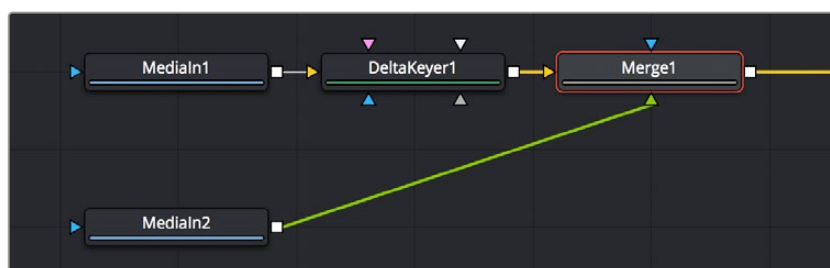
A MediaIn node branched to two different kinds of inputs, used separately.

Connecting Merge Nodes

The Merge node is the primary tool available for compositing images together. Each Merge node is capable of combining two inputs to create a third, using standard compositing methods and composite modes.

For more extensive information about the Merge node, see *Chapter 105, “I/O Nodes,”* in the DaVinci Resolve Reference Manual or Chapter 45 in the Fusion Reference Manual.

For this chapter, all you need to know is that if you attach a background image to the Background input (such as a landscape), and a foreground image with an alpha channel to the Foreground input (such as a graphic with an alpha channel), the Merge node will combine them into a single image for further compositing.



Two MediaIn nodes and a DeltaKeyer node attached to a Merge node, creating a composite.

Each Merge node has three inputs:

- **Background (orange):** The default input. Whichever image is connected to this input defines the output resolution of the Merge node.
- **Foreground (green):** The secondary input, meant for whichever image you want to be “on top.”
- **Effect Mask (blue):** An optional input you can use to attach a mask or matte with which to limit the effect of the Merge node.

It's important to make sure you're attaching the correct nodes to the correct inputs to ensure you're getting the result you want, and it's important to keep these inputs in mind when you connect to a Merge node. Of course, you can always drag a connection to a specific input to make sure you're connecting things the way you need. However, if you're in a hurry and you simply drag connections right on top of a Merge node:

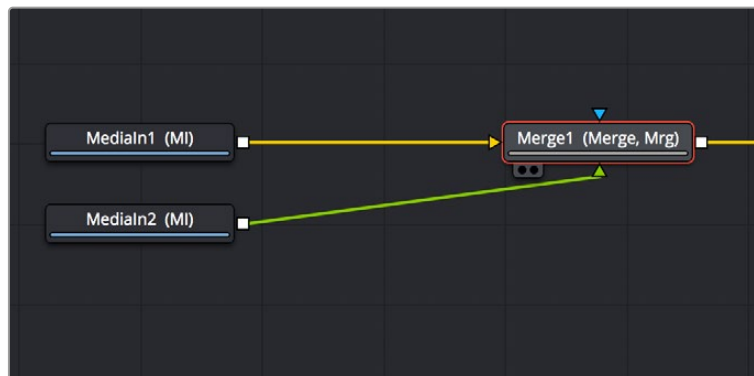
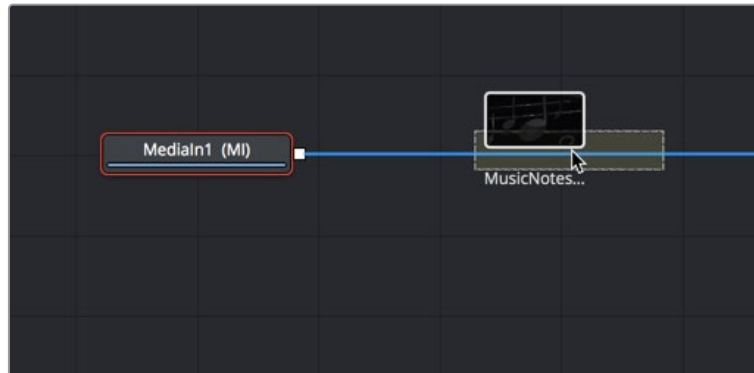
- The first connection will be made to the background input.
- The second connection will be made to the foreground input.
- The third connection will be made to the effect mask input.

TIP: When you add a Merge node after a selected node by clicking the Merge button on the toolbar, by clicking on the Merge icon in the Effects Library, or by right-clicking a node in the node tree and choosing Insert Tool > Composite > Merge from the contextual menu, the new Merge node is always added with the background connected to the upstream node coming before it.

Automatically Creating a Merge Node When Adding Nodes

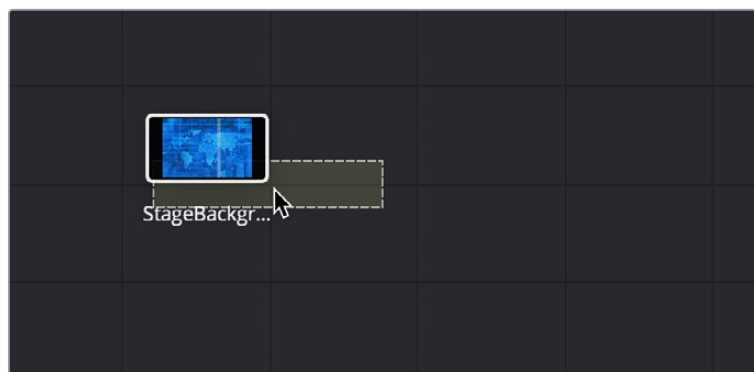
There's a nice shortcut for connecting Merge nodes if you want to connect the incoming clip immediately to your node tree as the top layer of a composite, and that's to drag a clip from an Operating System window or a Generator from the Effects Library right on top of any connection line.

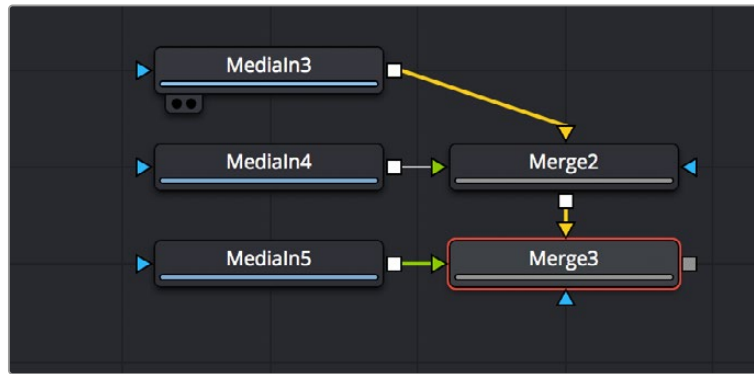
When you drop the resulting node, this automatically creates a Merge node, the background input of which is connected to the next node to the left of the connection you dropped the clip onto, and the foreground input of which is connected to the new node that represents the clip or Generator you've just added.



Dragging a node from the Media Pool onto a connection (top), and dropping it to create a Merge node composite (bottom).

Additionally, If you drag two or more nodes from an OS window into the Node Editor at the same time, Merge nodes will be automatically created to connect them all, making this a fast way to initially build a composite.

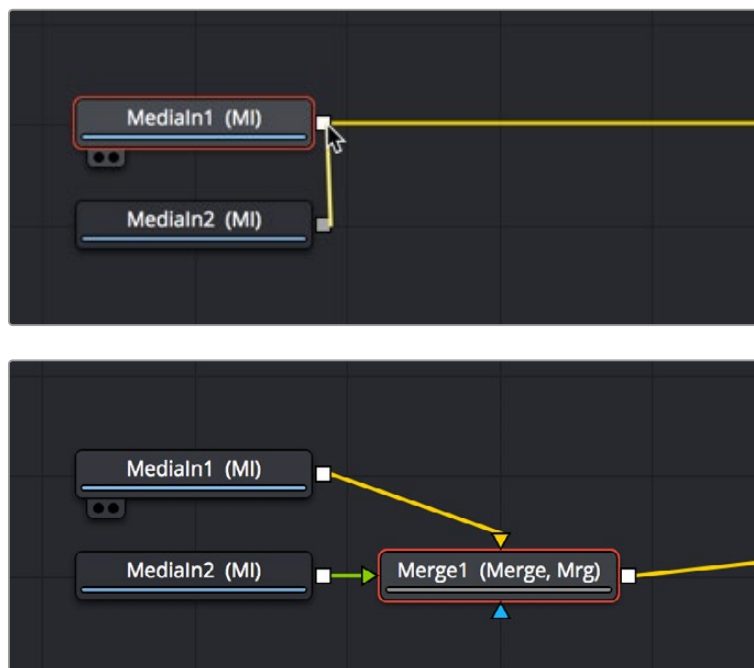




By dragging three nodes from an OS window to the Node Editor (top), Merge nodes are automatically created to connect them all (bottom).

Automatically Creating a Merge Node by Connecting Two Outputs

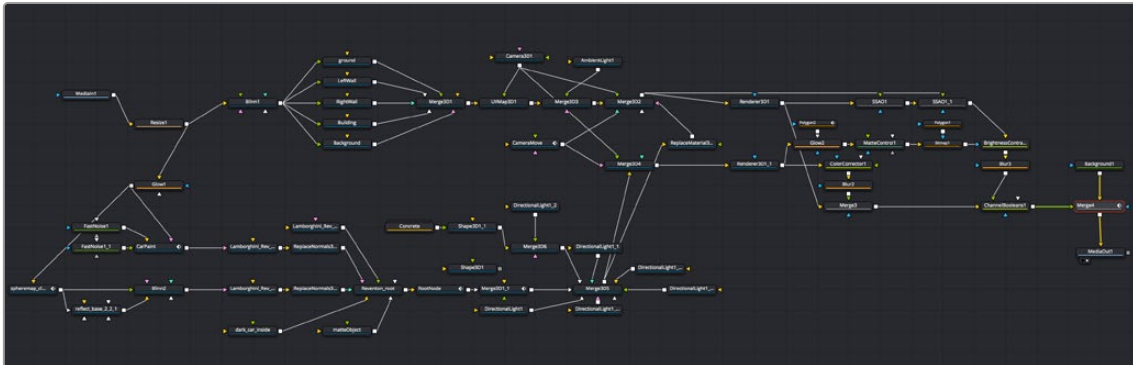
Here's an endlessly useful shortcut for when you have a disconnected node that you want to composite over another node. Drag a connection from the output of the node you want to be the foreground layer, and drop it on top of the output of the node you want to be the background layer, and a Merge node will be automatically created to build that composite.



Dragging a connection from a disconnected node to another node's output (top), and dropping it to create a Merge node composite (bottom).

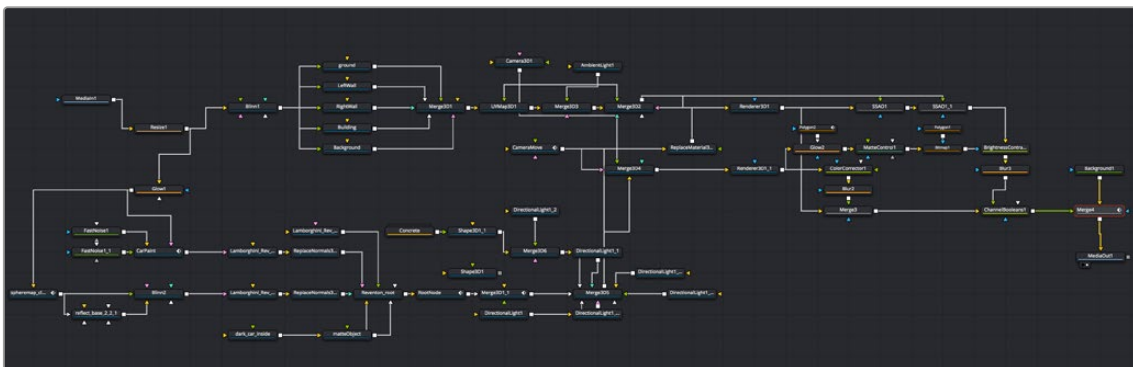
Connection Options and Routers

By default, the Node Editor uses linear connections that are drawn straight between any two connected nodes. While efficient, this sometimes causes connection lines to overlap nodes, which some people feel interferes with the view of the Node Editor.



Linear connections between nodes.

If you like, you can change how connections are drawn by enabling *orthogonal* connections, which automatically draws lines with right angles to avoid having connections overlap nodes.



Optional orthogonal connections between nodes.

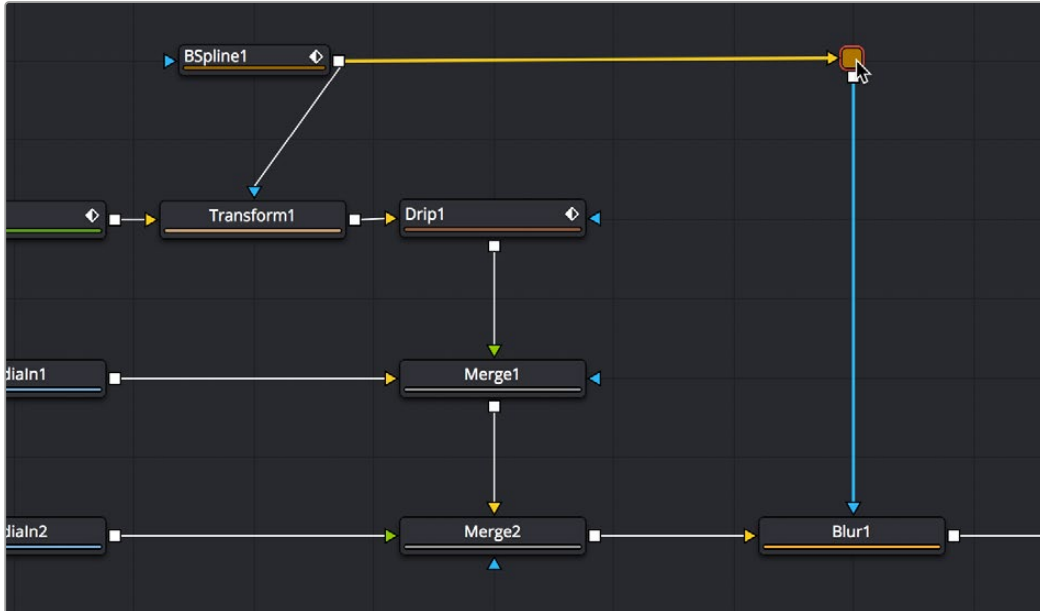
Functionally, there's no difference to your composition; this only affects how your node tree appears.

To change how connections are drawn in the Node Editor:

- Right-click the Node Editor background and choose one of the following from the contextual menu.
- Options > Direct Pipes
- Options > Orthogonal Pipes

Using Routers to Reshape and Branch Connections

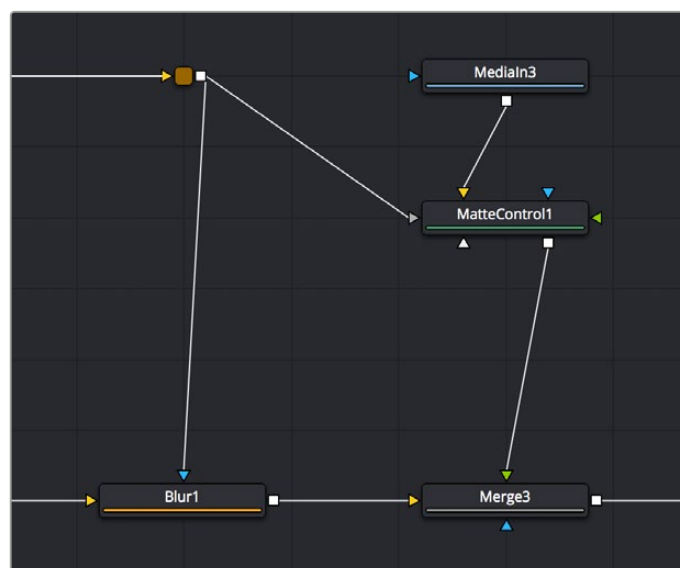
If you want to force a particular connection to be drawn at an angle to keep your node tree tidy, you can add a router to either linear or orthogonal connections to force an angle so it will be drawn however you like.



A router added to force a connection to be drawn at an angle.

Routers are tiny nodes with a single input and an output, but with no parameters except for a comments field (available in the Inspector), which you can use to add notes about what's happening in that part of the composition.

Even more usefully, you can branch a router's output to multiple nodes, which makes routers even more useful for keeping node trees neat in situations where you want to branch the output of a node in one part of your node tree to other nodes that are all the way on the opposite end of that same node tree.



A router branching its output to multiple nodes.

Methods of using routers:

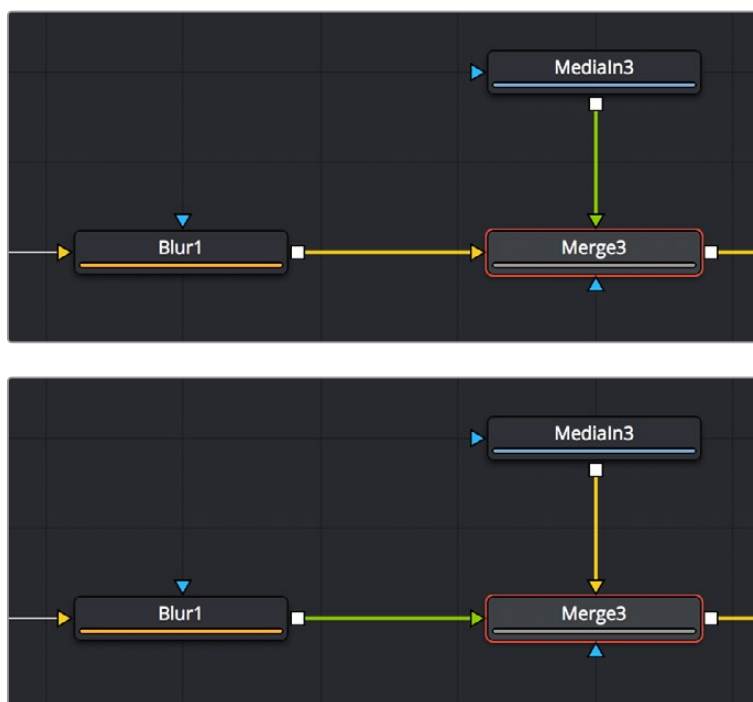
- **To add a router to a connection:** Option-click anywhere on a connection.
- **To move a router:** Drag the router to a new location, and the connection will reshape itself as necessary.
- **To branch a router's output:** Drag a connection from the router output to the input of another node. You can branch a router's output as many times as you need to.
- **To remove a router:** Select any router and press the Delete key, or right-click a router and choose Delete from the contextual menu.

Swapping Node Inputs

For multiple-input nodes such as the Merge, Merge 3D, and Dissolve nodes, there's a quick method of swapping the Primary and Secondary inputs, such as the foreground and background inputs of a Merge tool, when you find you've accidentally connected them in the wrong order. If a node has more than two of its inputs connected, only the foreground and background inputs will be swapped.

To swap the primary inputs of a multi-input node, do one of the following:

- Select a node and press Command-T to reverse its inputs.
- Right-click a node and choose Swap Inputs from the contextual menu.



Before swapping node inputs (top), and after swapping node inputs (bottom), the connections don't move but the colors change.

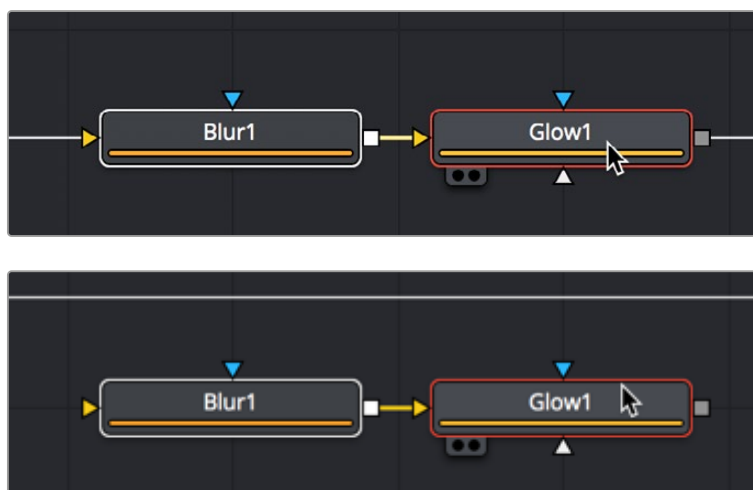
Inputs can move freely around the node, so swapping two inputs doesn't move the connection lines; instead, the inputs change color to indicate you've reversed the background (orange) and foreground (green) connections.

Extracting and Inserting Nodes

When building a composition, you'll often find that you need to rearrange nodes that you've already added, in order to connect them in different ways to obtain a better result. Happily, this is easy to do by extracting one or more nodes from one part of a node tree and inserting them at another part of the node tree.

To extract one or more nodes from their position in the node tree:

- **To extract a single node:** Hold down the Shift key, drag a node from the node tree up or down to disconnect it, and then drop the node before releasing the Shift key. That node is now detached, and the output of the next upstream node is automatically connected to the input of the next downstream node to fill the gap in the node tree.
- **To extract multiple nodes:** Select the nodes you want to extract, hold down the Shift key, drag one of the selected nodes up or down to disconnect them, and then drop the node before releasing the Shift key. Those nodes are now detached (although they remain connected to one another), and the output of the next upstream node is automatically connected to the input of the next downstream node to fill the gap in the node tree.

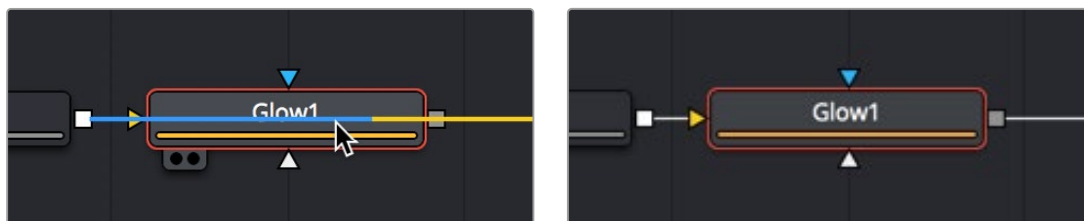


Before extracting a pair of nodes (left), and after extracting a pair of nodes (right).

After you've extracted a node, you can re-insert it into another connection somewhere else. You can only insert one node at a time.

To insert a disconnected node in the Node Editor between two compatible nodes:

- 1 Hold down the Shift key and drag a disconnected node directly over a connection between two other nodes.
- 2 Once the connection highlights, drop the node, and then release the Shift key. That node is now attached to the nodes coming before and after it.



Before inserting a node (left), and after inserting a node (right).

TIP: If you hold down the Shift key, you can extract a node and re-insert it somewhere else with a single drag.

Cut, Copy, and Paste Nodes

The standard operations of cut, copy, and paste are also available in the Node Editor. You can use them to temporarily remove nodes from the Node Editor, create duplicate nodes, or even copy the settings from one node and paste those settings into another node with compatible settings.

Cut, Copy, and Paste in the Node Editor

The standard commands all work, but with some special features specific to the Node Editor.

To copy one or more selected nodes, do one of the following:

- Right-click a node and choose Copy from the contextual menu.
- Choose Edit > Copy from the Edit menu (Command-C).

To cut one or more selected nodes, do one of the following:

- Right-click over the node and choose Cut from the contextual menu.
- Choose Edit > Cut from the Edit menu (Command-X).

When you paste into the Node Editor, you create a copy of the last node or nodes you've cut or copied. When pasting, there are a few different things you can do to control where pasted nodes appear.

To paste one or more selected nodes, do one of the following:

- **To paste nodes to be inserted after another node:** Select the node in the node tree you want to insert the pasted node(s) to, and choose Edit > Paste (Command-V).
- **To paste nodes to be disconnected from the rest of the node tree:** Deselect all nodes, and then choose Edit > Paste (Command-V), or right-click anywhere in the Node Editor and choose Paste from the contextual menu.
- **To paste disconnected nodes in a specific area of the Node Editor:** Deselect all nodes, and then click the place in the Node Editor where you want pasted node(s) to appear, and choose Edit > Paste (Command-V), or right-click anywhere in the Node Editor and choose Paste from the contextual menu.
- **To paste a node to replace an existing node in the Node Editor:** Right-click a node in the Node Editor that you want to replace, choose Paste from the contextual menu, and when a dialog appears asking if you want to replace that node, click OK. This only works when you use the contextual menu command.

TIP: When you paste a MediaIn, Loader, or Generator node so it will be inserted after a selected node in the node tree, a Merge tool is automatically created and used to composite the pasted node by connecting it to the foreground input. While this can save you a few steps, some artists may prefer to perform these sorts of merges manually, so this can be changed using the Defaults panel in the Auto tools section of the Fusion > Fusion Settings.

Pasting Node Settings

Instead of pasting a node, you can choose to paste just the parameter settings that you copied from another node. This can be useful if you've carefully set or animated parameters in one node that you want to also use in another node.

Note that you can paste settings between two nodes of the same type, or between two entirely different kinds of nodes that happen to have one or more of the same parameters in the Inspector. When copying settings from one type of node to another, only the settings that match between two nodes will be copied. A common example is to copy an animated Center parameter from a Transform node to the Center parameter of a Mask node.

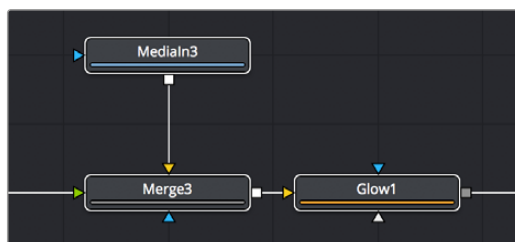
To Paste settings from one node to another:

- 1 Select a node that has settings you want to copy, and choose Copy from the Edit menu (Command-C).
- 2 Right-click a node you want to paste those settings to, and choose Paste Settings from the contextual menu.

Copying and Pasting Nodes to and from Any Text Editor

The format of nodes in the Node Editor is not binary, but is in fact a simple text format. The implications of that may not be obvious, but one example benefit is clear when you start dealing with nodes.

One or more nodes can be copied from the Node Editor and pasted directly into a text editor or email. This pastes the selection in text format, just as it's saved internally in Fusion. For example, if you copy the following set of three nodes:



A set of three nodes being copied.

And you then paste into a new text editing document, you get the following:

```

Tools = ordered() {
  MediaIn3 = MediaIn {
    ExtentSet = true,
    CustomData = {
      MediaProps = {
        MEDIA_HEIGHT = 1080,
        MEDIA_MARK_IN = 0,
        MEDIA_MARK_OUT = 244,
        MEDIA_NAME = "868838838-serial-abstract-waves-crystal-PRORESHD1080.mov",
        MEDIA_NUM_FRAMES = 245,
        MEDIA_NUM_LAYERS = 1,
        MEDIA_PAN = 1,
        MEDIA_PATH = "/Volumes/Media Raid Too/DaVinci Media/Fusion Manual Example Media/Pond5 Media/Textures/868838838-serial-abstract-waves-crystal-PRORESHD1080.mov",
        MEDIA_SRC_FRAME_RATE = 25,
        MEDIA_START_FRAME = 8,
        MEDIA_WIDTH = 1920
      },
    },
    Inputs = {
      GlobalOut = Input { Value = 244, },
      MediaID = Input { Value = "4888cadC-2c6f-44a8-be08-52f29b48f6d5", },
      Layer = Input { Value = "", },
      ClipTimeEnd = Input { Value = 244, },
    },
    ViewInfo = OperatorInfo { Pos = { 825, 115.5 }, },
  },
  Merge3 = Merge {
    Inputs = {
      Background = Input {
        SourceOp = "MediaIn3",
        Source = "Output",
      },
      PerformDepthMerge = Input { Value = 0, },
    },
    ViewInfo = OperatorInfo { Pos = { 825, 181.5 }, },
  },
  Glow1 = Glow {
    CtrlWZoom = false,
    Inputs = {
      Blend = Input { Value = 0.2, },
    },
    ViewInfo = OperatorInfo { Pos = { 935, 181.5 }, },
  },
}
}

```

The same three nodes pasted into a text editor.

At this point, you have the option of editing the text (if you know what you're doing), emailing it to colleagues, or storing it in a digital notepad of some sort for future use. To use this script in Fusion again, you need only copy it and paste it back into the Node Editor.

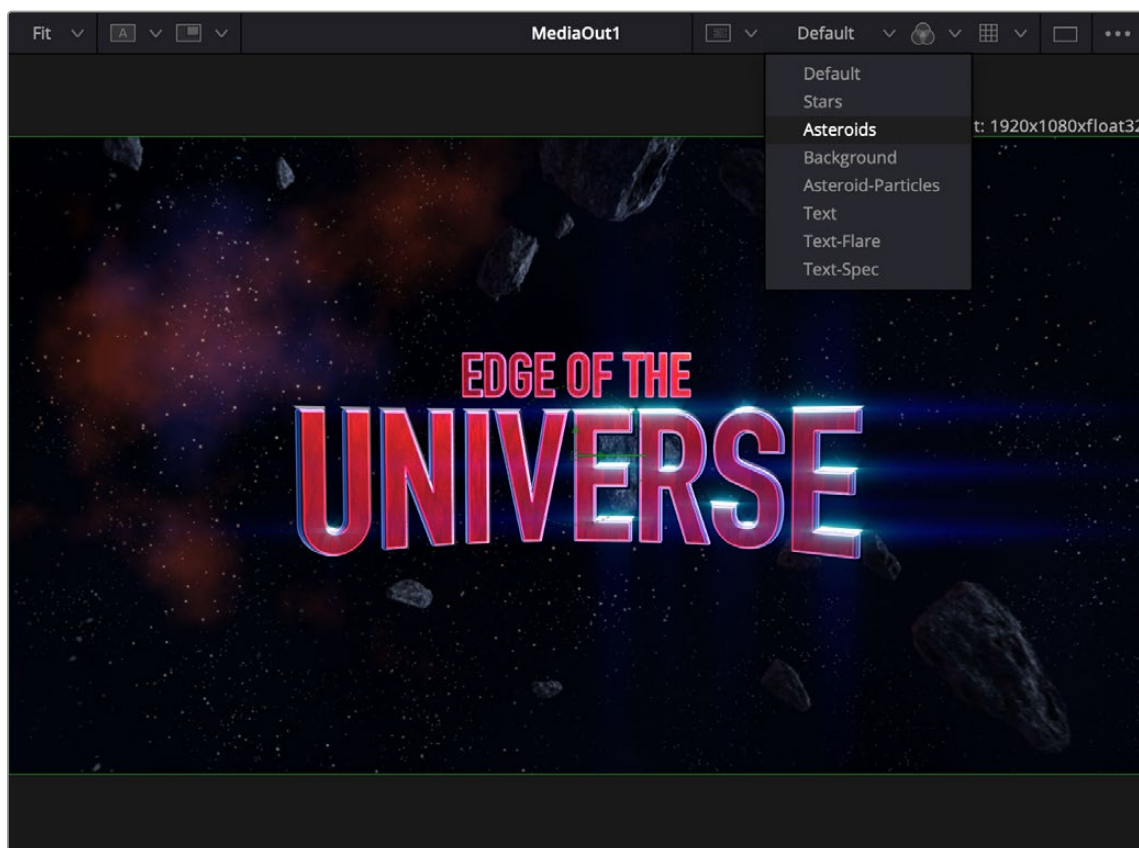
TIP: This is a very easy way to pass specific node settings back and forth between artists who may not be in the same room, city, or country.

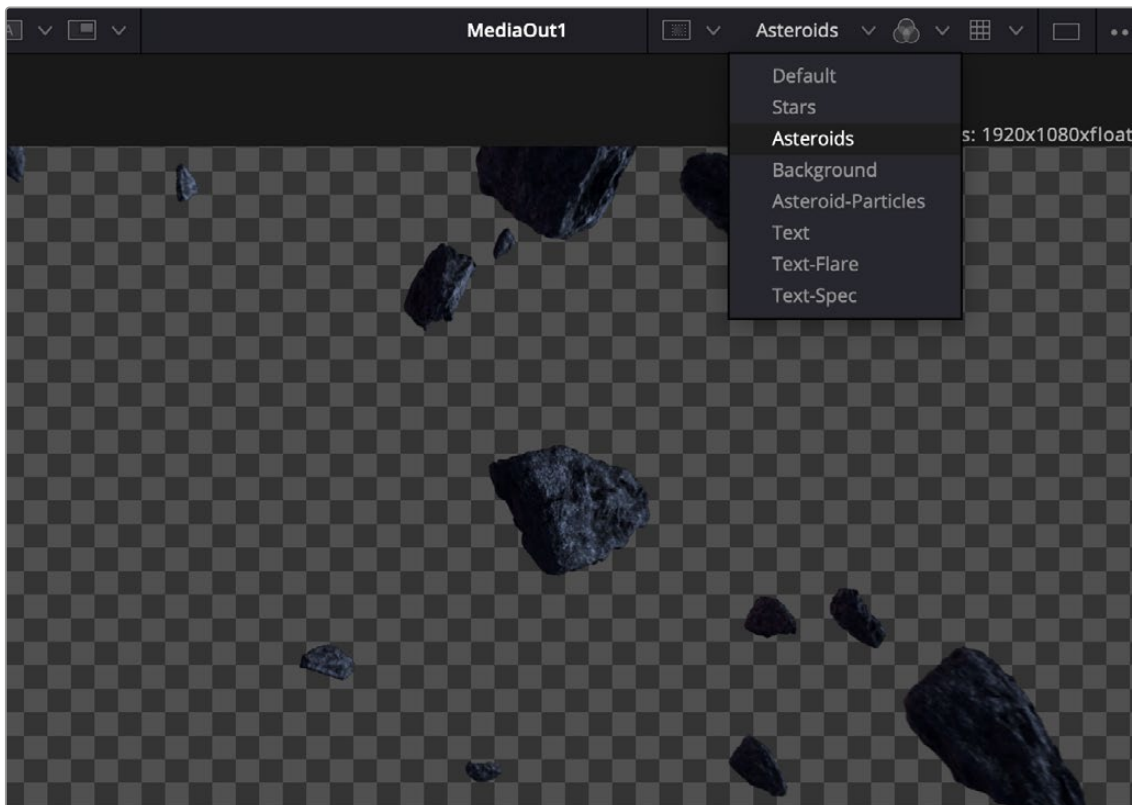
Using Multilayer Nodes (Studio Version Only)

Fusion can intelligently handle multilayer image workflows with support for EXR, Stereoscopic 3D, and PSD assets. Users can see a list of layers on the Fusion viewers and interact with layers in each node in the Settings tab in the Inspector. As this layer structure is retained throughout the toolset, this means you can isolate a tool to work on an individual layer, while still passing through the other layers unmodified.

Viewer

When loading a multilayer file, you can preview specific layers in the Viewer and select a layer from the drop-down list. Files without layer support will show only a 'default' option from the drop-down menu.



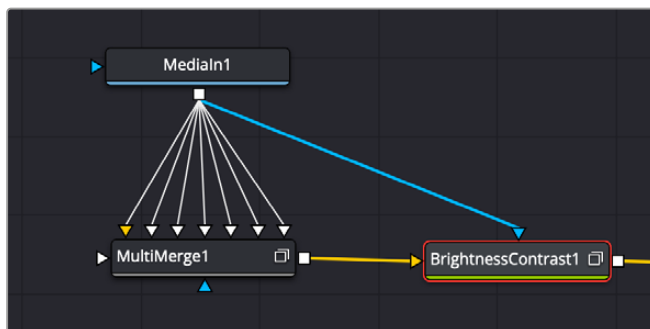


Using the Layer drop-down in the Viewer to select just the Asteroids layer from a multilayer file for viewing

Nodes and Inspector

In the Settings tab you'll find a Layer category where you have access to a number of controls to affect certain layers and to control layers from each tool input.

- **Process Layers:** Select the layer to process for effects from:
 - Default Layer (default):** To apply the effect on the base/unnamed layer, not affecting other layers.
 - All Layers:** To apply the effect on all available layers.
 - Custom:** Choice of other available layers to apply the effect on, not affecting other layers.
- **Input Layer:** The Input Layer lists all layers available from the upstream node of the default input. The selected layer can be used by the current tool or further down the node tree in isolation.
- **Effect Mask Layer:** If the tool has an Effect Mask input, this control lists the available layers that can provide mattes for the tool.



Connecting the multilayer file, both directly to the BrightnessContrast Effect Mask (blue arrow) in addition to its input, allows you to quickly create perfect masks of any layer in the file.

Tools with other named inputs (like **Image**) can show additional layer controls with a corresponding name (like **Image Layer**).

Input Layer Controls

For optimal performance, Fusion tools request only the processed layers from each input. The Input Layer, Effect Mask Layer, Image Layer and other input layer controls allow the user to customize this behavior. In each input control, you can select Auto or Match or select an explicit layer value to map:

- **Auto (Default):** Requests the layer selected in the Process Layers selection. If the layer does not exist, the tool falls back to the default layer.
- **Match:** Requests the layer selected in the Process Layers selection. If the layer does not exist, the tool shows a failure.
- **Explicit Layer Selection:** Requests the explicit layer selection from one of the listed layers. This isolates the specific layer for use in the current tool or further down the node tree.

For Fusion plugins, the Process Layer and Input Layer controls are automatically generated in the Inspector's Settings tab. Creators of plugins can decide to define and place the controls explicitly in another place, similar to how the MultiMerge places the controls in the main controls section.

Some Fusion tools include controls to handle additional layer-based decisions.

For example:

- The Renderer3D shows a new Eye control > Layers option when using a stereoscopic camera, allowing the left and right eyes to be output as individual layers.
- When combining multilayered sources using composite nodes (e.g., MultiMerge), Channel Booleans, or a MatteControl, the Settings tab has additional controls to select Background or Foreground options to isolate and combine the selected layers.
- The Combiner tool can be used to create custom layers. Set the Mode to Layer. This exposes layer name fields allowing you to name the new layers.

Instancing Nodes

Normally, when you use copy and paste to create a duplicate of a node, the new node is completely independent from the original node, so that changes made to one aren't rippled to the other. However, there are times when two nodes must have identical settings at all times. For example, when you're making identical color corrections to two or more images, you don't want to constantly have to adjust one color correction node and then manually adjust the other to match. It's a hassle, and you risk forgetting to keep them in sync if you're working in a hurry.

While there are ways to publish controls in one node and connect them to matching controls in another node, this becomes prohibitively complex and time consuming for nodes in which you're making adjustments to several controls. In these cases, creating "instanced" nodes is a real time-saver, as well as an obvious visual cue in your node tree as to what's going on.

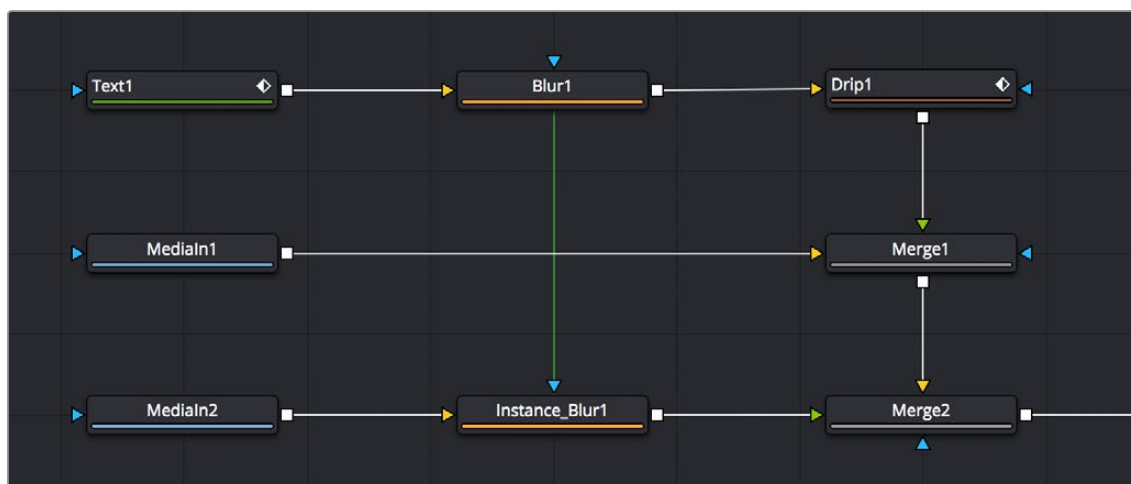
Using Instanced Nodes

Instanced nodes are nodes that have been created using the Paste Instance command, and which share settings with the original node so that a change made to one instanced node is also automatically applied to all other instances of that node (as well as the original node you copied).

To create an Instance, do the following:

- 1 Select a node you want to instance, and copy it (Command-C).
- 2 Do one of the following:
 - **To create a disconnected instance of a node:** Right-click in the background of the Node Editor, and choose Paste Instance from the contextual menu (Command-Shift-V).
 - **To insert an instanced node between two other nodes:** Select a node that's upstream of where you want to insert the instanced node, and press Command-Shift-V. Alternatively, you can right-click directly on a connection line, and choose Paste Instance from the contextual menu.

However you paste an instance, the name of that instanced node takes the form "Instance_NameOfNode." If you paste multiple instances, each instance is numbered "Instance_NameOfNode_01."



A green link line shows an instanced Blur node's relationship to the original Blur node it was copied from.

When a node tree contains instanced nodes, a green line shows the link between the original node and its instances. You have the option to hide these green link lines to reduce visual clutter in the Node Editor.

To toggle the visibility of green instance link lines in the Node Editor:

- 1 Right-click anywhere in the background of the Node Editor.
- 2 Choose Options > Show Instance Links from the contextual menu.

If you've been using an instance of a node and you later discover you need to use it to apply separate adjustments, you can "de-instance" the node.

To de-instance a node, making it independent:

- 1 Right-click an instanced node.
- 2 Choose Deinstance from the contextual menu. That node is now independent from the original node. Once you de-instance a node, you cannot re-instance it, but you can undo the operation.

NOTE: If you've de-instanced a node and you cannot undo the operation because you've restarted DaVinci Resolve, you can only recreate an instance by copying the original and pasting an instance again.

De-Instancing and Re-Instancing Specific Parameters

By default, every parameter in an instanced node is linked to the original node, so that any change you make is rippled across. However, from time to time you'll find the need to independently adjust just one or two parameters while keeping the rest of that node's parameters linked. For this reason, instead of de-instancing the entire tool, you can de-instance individual parameters.

To de-instance a single parameter:

- Right-click on a parameter's name or value in the Inspector, and choose Deinstance from the contextual menu.

If you've only de-instanced individual parameters, you can re-instance those parameters later on if you change your mind.

To re-instance a single parameter:

Right-click on a parameter's name or value in the Inspector, and choose Reinstance from the contextual menu. That parameter immediately inherits the setting of the original node.

Keeping Node Trees Organized

Similar to working with files on your desktop, even the simplest of composites require you to do some amount of organization. In this section we'll look at some basic node operations, some of which you may already be familiar with just from using your computer's operating system or other applications.

Moving Nodes

Selecting one or more nodes and dragging them moves them to a new location, which is one of the simplest ways of organizing a node tree, by grouping nodes spatially according to the role they play in the overall composition.

Keep in mind that the location of nodes in the Node Editor is purely aesthetic, and does nothing to impact the output of a composition. Node tree organization is purely for your own peace of mind, as well as that of your collaborators.

TIP: Once you've arranged the nodes in a composition in some rational way, you can use the Sticky Note and Underlay tools to add information about what's going on and to visually associate collections of nodes more definitively. These tools are covered later in this section.

Snapping Nodes to the Grid

By default, you can position nodes freely wherever you want them to be. However, keeping nodes and connection lines straight and aligned can make them easier to read. To help keep them aligned, you can have nodes you're dragging automatically snap to the grid.

To have nodes snap to the grid as they're dragged:

- Right-click over an empty area of the Node Editor, and choose Arrange Tools > To Grid from the contextual menu. All nodes you drag now snap to the nearest grid coordinate.
- Right-click over an empty area of the Node Editor, and choose Arrange Tools > To Connected from the contextual menu. All nodes you drag now snap to the horizontal or vertical position of the nodes they're attached to.

TIP: You can set “Arrange to Grid” or “Arrange to Connected” as the default for new compositions by choosing Fusion > Fusion Settings in DaVinci Resolve or File > Preferences in Fusion Studio, and turning the Fusion > Node Editor > Arrange To Grid or Arrange to Connected checkboxes on.

Commands to “Clean Up” a Node Tree

The grid in the background of the Node Editor can be used to align nodes, either by eye or automatically.

To “clean up” an unruly node tree:

Right-click in an empty section of the Node Editor, and choose Line Up All Tools to Grid from the contextual menu. All nodes in the Node Editor will move to align and center themselves along the nearest grid lines.

To “clean up” only one or more selected nodes:

Right-click one of the selected nodes and choose Line Up to Grid from the contextual menu. All selected nodes will move to align and center themselves along the nearest grid lines, while all unselected nodes will be left as they are.

Renaming Nodes

Each node that's created is automatically assigned a name (based on its function) and a number (based on how many of that type of node have been created already). For example, the first Blur node added to a composition will be called Blur1, the second will be Blur2, and so on. Although initially helpful, larger compositions may benefit from important nodes having more descriptive names to make it easier to identify what they're actually doing, or to make it easier to reference those nodes in expressions.

To rename a node:

- 1 Do one of the following:
 - Right-click a node and choose Rename from the contextual menu.
 - Select a node and press F2.
- 2 When the Rename dialog appears, type a new name, and then click OK or press Return.

NOTE: If multiple nodes are selected, multiple dialogs will appear asking for a name for each tool.

Since Fusion can be scripted and use expressions, the names of nodes must adhere to a scriptable syntax. Only use alphanumeric characters (no special characters), and do not use any spaces.

Also, you cannot start a node name with a number. If you accidentally create a name that doesn't exactly follow the guidelines, spaces and invalid characters will be automatically deleted.

If you want to see the original node types instead of the node names, press and hold Command-Shift-E.

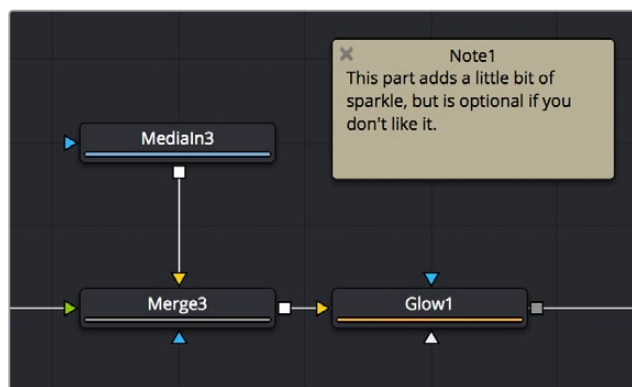
Changing Node Colors

You can change the color of any node by selecting it, opening the Inspector, and choosing a new color from the Node Color pop-up in the Inspector header for that node. Alternatively, you can right-click a node and choose a color from the Set Color submenu.

To return a node to its regular color, right-click it and choose Set Color > Clear Color from the contextual menu, or open the Node Color pop-up for a node in the Inspector, and choose Clear Color.

Using Sticky Notes

A good way to add notes about different parts of a composition, client feedback about various details, and other information you want to keep track of, is to add Sticky Notes to the Node Editor.



A Sticky Note in the Node Editor.

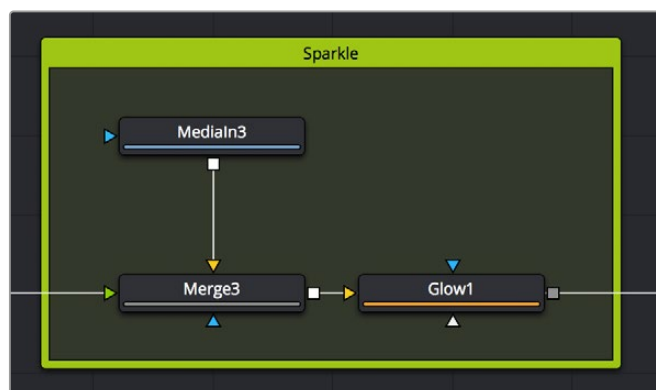
Sticky Notes are yellow boxes in which you can type whatever text you want. They can be resized, moved, and collapsed when they're not being edited, but once created they remain attached to the background of the Node Editor where you placed them until you either move them or delete them.

Methods of working with Sticky Notes:

- **To create a Sticky Note:** Click somewhere in the Node Editor where you want a Sticky Note to appear. Then, press Shift-Spacebar, type sticky, and press the Return key when the Sticky Note appears in the Select Tool window. Alternatively, you can open the Effects Library, open the Tools > Node Editor category, and click or drag the Sticky Notes node to create a new one.
- **To open a Sticky Note to full size:** Double-click a minimized Sticky Note and it expands to a larger, resizable yellow box.
- **To edit a Sticky Note:** If necessary, double-click a Sticky Note to open it to full size, and then click once in the body of the note to place a text cursor. You can edit text within the Sticky Note just like any other text editor.
- **To rename a Sticky Note:** Right-click a Sticky Note, choose Rename, type a new name into the Rename dialog, and click OK. Alternatively, you can select a Sticky Note, press F2 to open the Rename dialog, and press Return to close it when you're done.
- **To resize a Sticky Note:** Double-click a Sticky Note to open it to full size, and then drag any of the edges or corners to make it larger or smaller.
- **To minimize a Sticky Note:** Click the close box at the upper left-hand corner of the Sticky Note, and it collapses to a small tile.
- **To delete a Sticky Note:** Right-click any Sticky Note and choose Delete from the contextual menu or select the Sticky note in the Node Editor and press the Delete key.

Using Underlay Boxes

Underlay Boxes are a good way of associating a collection of nodes that work together to perform a specific task in your composition. They're simply colored rectangles that you can put nodes inside of. Once you place nodes inside an Underlay, you can move the Underlay, and all the nodes within move along with it.



An Underlay in the Node Editor.

Underlay Boxes can be named to identify the purpose of that collection of nodes, and they can be colored to be distinct from other Underlay Boxes or to adhere to some sort of color code for your compositions.

Methods of working with Underlay Boxes:

- **To create an Underlay Box:** Click somewhere in the Node Editor where you want the Underlay Box to appear. Then, press Shift-Spacebar, type under, and press the Return key when the Underlay Box appears in the Select Tool window. Alternatively, you can open the Effects Library, open the Tools > Node Editor category, and click or drag the Underlay Box node to create a new one.

- **To create an Underlay Box around specific nodes:** Select the nodes in the Node Editor that you want surrounded by an Underlay Box. Then, press Shift-Spacebar, type under, and press the Return key when the Underlay Box appears in the Select Tool window. Alternatively, you can open the Effects Library, open the Tools > Node Editor category, and click the Underlay Box node to have it added and sized to encompass all the selected nodes.
- **To resize an Underlay Box:** Drag any of the edges or corners to make it larger or smaller.
- **To rename an Underlay Box:** Option-click the Underlay Box to select just the box and not the contents, and then right-click it and choose Rename (or press F2). Type a new name into the Rename dialog and click OK or press Return.
- **To change the color of an Underlay Box:** Option-click the Underlay Box to select just the box and not the contents, and then right-click it and choose a color from the Set Color submenu.
- **To put nodes inside of an Underlay Box:** Select the nodes you want to place inside an Underlay Box, and then drag them to fit inside. The Underlay Box must be big enough to fit all the nodes. Alternatively, you can place an Underlay Box near a collection of nodes you want to put inside it, and then resize the Underlay Box to encompass all those nodes.
- **To move an Underlay Box and all its nodes:** Once nodes have been placed inside an Underlay Box and have been deselected, you can move the entire collection of nodes together by dragging the Underlay Box by its title bar.
- **To remove nodes from an Underlay Box:** There are two ways you can remove nodes from an Underlay Box.

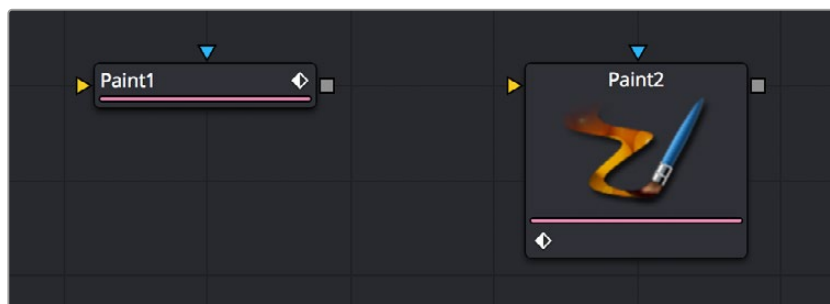
With both the Underlay Box and nodes deselected, drag a bounding box or Command-click to select all nodes in the box you want to remove, and drag them out.

Resize the Underlay Box so that it's smaller than the collection of nodes it originally encompassed. Once an Underlay Box is so small that even the last node sticks out beyond its edge, those nodes are automatically removed from the Underlay Box, and you can move or delete the Underlay Box without moving those nodes.

- **To delete an Underlay Box and all nodes within:** Select an Underlay Box and press the Delete key to delete both the Underlay Box and all nodes found inside it. If you don't also want to delete the nodes, first drag the nodes out of the box.
- **To delete an Underlay Box but keep all nodes within:** Option-click the Underlay Box to select it and not the nodes, and then press the Delete key. The nodes within remain where they were.

Node Thumbnails

Once a source or an effect has been added to the Node Editor, it's represented by a node. By default, nodes are rectangular and thin, making it easier to fit reasonably complicated grades within a relatively small area. However, if you like, you can also display node thumbnails.



A node in the Node Editor shown without and with a thumbnail.

Nodes can be displayed as a small rectangle or as a larger square. The rectangular form displays the node's name in the center, while the square form shows either the tool's icon or a thumbnail of the image it is outputting.

TIP: Even if you're not displaying node thumbnails, you can quickly obtain detailed information about a node and the data it's processing by hovering your pointer over it in the Node Editor and viewing the tooltip bar below.

Choosing Which Nodes Show Thumbnails

If you want to use node thumbnails to help visually identify media and operations in your node trees, there are a variety of options for which nodes should display thumbnails in the contextual menu that appears when you right-click anywhere in the background of the Node Editor.

Force All Tile Pictures

This option shows thumbnails for every single node in the Node Editor. This can make simple node trees easier to read, but it'll make all node trees take up considerably more room.

NOTE: If Show Thumbnails is enabled, nodes may not update until the playhead is moved in the Time Ruler.

Force Active Tile Pictures

You may also choose to only show thumbnails for nodes that are currently selected, which can make it easier to see which node you're working on. When nodes become deselected, the thumbnails will be hidden again.

Force Source Tile Pictures

This enables thumbnails for all MediaIn and Loader nodes in the Node Editor, as well as all generators, and is a great way to be able to quickly see where all the clips are in a composition.

Force Mask Tile Pictures

This enables thumbnails for all Mask nodes in a composition, which can make them easier to distinguish when you're building complex shapes made from multiple Mask nodes.

Manually Showing Tile Pictures and Node Options

You also have the option of manually choosing which nodes you'd like to show thumbnails. For example, there may be certain key points of the node tree where you'd like to see a small visual representation of what's happening in the composition.

To toggle thumbnails for one or more specific nodes:

- 1 Select one or more nodes in the Node Editor.
- 2 Right-click one of the selected nodes, and choose one of the following from the contextual menu:
 - Show > Show Tile Pictures
 - Show > Show Modes/Options

When you've manually enabled thumbnails for different nodes, they'll remain visible whether or not those nodes are selected.

Switching Thumbnails between Images and Icons

Whenever you enable node thumbnails, you have the choice of having these thumbnails either display an image of the state of the image at that node, or you can instead choose to display the icon for that particular node. The setting for this affects all nodes at once.

To display icons instead of thumbnails:

Right-click anywhere in the background of the Node Editor and deselect Show Thumbnails in the contextual menu.

Sometimes Nodes Only Show Icons

As you add more and more nodes to a composition, you'll notice that some nodes never display an image in their thumbnail. In these cases, the default icon for that node is displayed instead of an image.

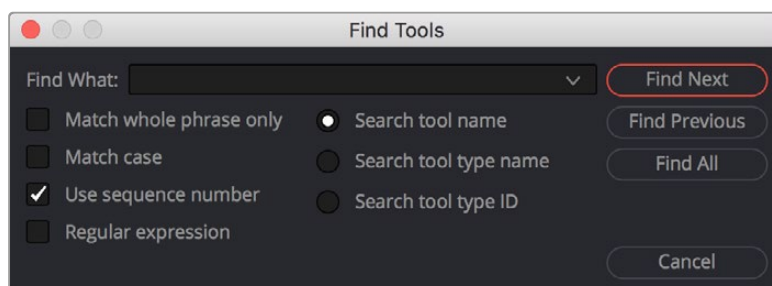
Most nodes in the Particle and 3D categories fall into this group. The exceptions are the pRender node and the Render 3D node. These two nodes are capable of displaying a rendered thumbnail if you have the menu options set for Thumbnails to be displayed.

In other cases, whether nodes display images in their thumbnail is more situational. Some Transform nodes are able to concatenate their results with one another, passing the actual processing downstream to another node later in the node tree. In this case, upstream Transform nodes don't actually process the image, so they don't produce a thumbnail.

In other situations where the Loader is not reading in a clip or the clip is trimmed in the Keyframes Editor to be out of range, it can cause the node not to process the image, so it will not produce a rendered Thumbnail image. Also, nodes that have been set to Pass Through mode are disabled and do not display a rendered Thumbnail image.

Finding Nodes

Modern visual effects require detailed work that often results in compositions with hundreds of nodes. For such large node trees, finding things visually would have you panning around the Node Editor for a long, long time. Happily, you can quickly locate nodes in the Node Editor using the Find dialog.



The Find dialog lets you quickly locate nodes wherever they are in the Node Editor.

Performing Simple Searches

To do simple searches using node names is easy.

To search for a node in the Node Editor:

- 1 Press Command-F, or right-click in an empty area of the Node Editor and choose Find from the contextual menu.
- 2 When the Find dialog appears, do the following:
 - Enter a search term in the Find field.
 - Choose search options, such as whether to match the whole phrase in the Find field, whether to match the case, whether to use a sequence number, or whether to use a regular expression in the Find field.
 - Choose what to search. Options include tool name, tool type name, or tool type ID.
- 3 To perform the find, do one of the following:
 - Click Find Next to try to select a downstream node matching the criteria.
 - Click Find Previous to try to select an upstream node matching the criteria.
 - Click Find All to try to select all nodes in the Node Editor that match the criteria.

The Find window closes. If either the Find Next, Find Previous, or Find All operations are successful, the found node or nodes are selected. If not, a dialog appears letting you know that the string could not be found.

TIP: Finding all the nodes of a particular type can be very useful if you want, for example, to disable all Resize nodes. Find All will select all the nodes based on the search term, and you can temporarily disable them by pressing the shortcut for Bypass, Command-P.

Using Regular Expressions

If you need to do more complicated searches, you can turn on the Regular Expression checkbox, which lets you enter some simple expressions with which to create more complex find operations. Some useful examples of regular expressions that are valuable include the use of Character Sets.

Character Sets

Any characters typed between two brackets [] will be searched for. Here are some examples of character set searches that work in Fusion.

[a-z]

Finds: Every node using a lower caps letter

[a-d]

Finds: Every lower caps letter from a to d, and will find nodes with a, b, c, or d

[Tt]

Finds: Every node with an upper case T or a lower case t

[aeiou]

Finds: Every vowel

[0-9]

Finds: Every numeral

[5-7]

Finds: Every numeral from five to seven, and will find nodes numbered with 5, 6, or 7

Custom Node Settings

When a node is added to the Node Editor, its parameters are set to the default values for that type of node. If you find yourself constantly readjusting the parameters of a node to a preferred starting point as soon as it's added to the node tree, you can override the default node settings with your own custom settings.

To save new default settings for a particular type of node:

- 1 Create a new node.
- 2 Open the Inspector and customize that node's settings to the new defaults you want it to have.
- 3 Right-click that node in the Node Editor, or right-click that node's control header in the Inspector, and choose Settings > Save Default from the contextual menu.

TIP: You can also save six different settings for a node in the Node Editor using the Version buttons at the top of the Inspector. For more information, see *Chapter 70, "Editing Parameters in the Inspector,"* in the DaVinci Resolve Reference Manual or Chapter 8 in the Fusion Reference Manual.

Managing Saved Settings

Custom node default settings are saved to a folder on your hard drive that's based on the Path Map > Defaults preference in the Fusion Settings. This path is customizable for facilities where multiple compositing artists use a common set of facility defaults, stored somewhere commonly accessible.

The default paths are:

For macOS systems, this path defaults to: /UserName/Library/Application Support/Blackmagic Design/DaVinci Resolve/Fusion/Defaults.

For Windows systems, this path defaults to C: \Users\<username>\AppData\Roaming\Blackmagic Design\DaVinci Resolve\Fusion\Defaults.

For Linux systems, this path defaults to: ~/.fusion/BlackmagicDesign/DaVinci Resolve/Fusion/Defaults.

If you browse this directory, the settings for each node are saved using a name taking the form INTERNALNAME_PUBLICNAME.settings, where INTERNALNAME is the internal name of the Fusion tool, and PUBLICNAME is the name of the Node that's derived from the internal Fusion tool. For example, the default setting for a Blur node would be called Blur_Blur.setting. This naming convention is partly to ensure that third-party plugin nodes don't overwrite the defaults for built-in Fusion nodes that happen to have the same name.

Resetting Defaults

Even if you've created new default settings for new nodes, you can always reset individual parameters to the original default setting. In addition, it's easy to restore the original default settings for new nodes you create.

To reset a single parameter to the original default settings:

- 1 Create a new node.
- 2 Open the Inspector and customize a parameter to the new default value you want it to have.
- 3 Right-click that parameter in the Inspector, and choose Set to Default from the contextual menu.

To reset every parameter in a node to the original defaults, do one of the following:

- Right-click on the node and choose Settings > Reset Default.
- Right-click that node's control header in the Inspector, and choose Settings > Reset Default.
- Delete the .setting file from the Defaults folder.

NOTE: When you use the Settings > Reset Default command, the default .setting file is deleted. If you want to save a node's settings as alternate settings, you should use the Settings > Save As command.

Saving and Loading Alternate Node Settings

Once you change parameter values for a node using the Inspector, those values can also be saved as an alternate setting for that node, which can be reused at a later time.

To save alternate settings for a node:

- 1 Right-click on a tool, and then choose Settings > Save As from the contextual menu.
- 2 When the Save File dialog appears, enter a name for the Setting and save it to your hard drive. Unlike saved defaults, the .settings files can be saved anywhere on the file system. They do not need to be in the Default Settings folder.

To load a saved setting for one or more nodes:

- 1 Right-click a node and choose Settings > Load from the contextual menu.
- 2 Use the Open File dialog to select the settings you want to load into that node, and then click Open. Those settings are now applied to that node.

Adding Saved Settings from the File System

Saved settings in your File system can also be used to create new nodes by dragging the .setting file into the Node Editor from a standard file browser. Once dropped, that setting turns into a new node.

TIP: If you drop a setting directly onto a connection line, the new node will be inserted onto that connection.

Node Modes Including Disable and Lock

Right-clicking one or more nodes and opening the contextual menu reveals a series of commands in the Modes submenu, some with accompanying keyboard shortcuts, that let you set control visibility, disable, lock, update, and cache nodes.

- **Show Controls:** Sets whether that node reveals its parameters in the Inspector when it's selected and whether its onscreen controls appear in viewers. On by default.
- **Pass Through:** (Command-P) Identical to the toggle switch in the Inspector that turns nodes off and on. Disabled nodes are ignored as image data is passed from the next previous upstream node to the next downstream node. On by default.
- **Locked:** (Command-L) Identical to the lock button in the Inspector that prevents a node from being edited in the Inspector. Off by default.
- **Update:** (Command-U) On by default. While this option is enabled, all changes to the node will cause it to re-render. When Update is disabled, you can still change the node's parameters, but those changes will not process or update the image until Update is re-enabled. While disabled, the last processed image for that node will be displayed as a freeze frame. One example of when this is useful is when you have a large or processor-intensive composition (such as a particularly intense particle system), and disabling this option temporarily will let you quickly make several quick parameter adjustments to different nodes without forcing you to wait for the node tree to re-render after every adjustment. Another example of when this is useful is when you want to quickly see the effect of animated downstream nodes while keeping upstream nodes that are too processor-intensive to play in real time from rendering additional frames.
- **Force Cache:** When enabled, this node's output for the current frame has an extremely high cache priority, essentially forcing it to stay cached in memory. Off by default.

Toggleing any one of these node modes displays a badge within that node indicating its state.

Node Editor Options

Right-clicking in an empty area of the Node Editor will bring up the contextual menu and the Options submenu. The Options submenu contains several choices that can be used to customize how the Node Editor looks and behaves.

- **Pipes Always Visible:** Enabling this option causes a connection to cross over a node instead of beneath it, sometimes making it easier to follow the connection's path.
- **Show Hidden Pipes:** When enabled, the Inspector option to Hide Incoming Connections in every node is overridden and all connections are displayed in the Node Editor.
- **Aspect Correct Tile Pictures:** Aspect Correct Tile Pictures forces the display of thumbnails to be aspect corrected, which is slower but visually more accurate. This option is enabled by default.
- **Full Tile Render Indicators:** Enabling this option causes the thumbnail to flash green when rendering, which makes it easier to identify which node is processing in a large, complex node tree.
- **Show Grid:** This option can be used to enable or disable the Node Editor's background grid.
- **Show Instance Links:** When enabled, the Node Editor draws a green connection between an instanced node and its parent.
- **Auto Remove Routers:** If routers are disconnected from a tool, they are automatically deleted from the Node Editor. This option is enabled by default to eliminate the need to delete orphaned routers.

- **Show Navigator:** Enabling this option displays a small overview window of the entire node tree in the Node Editor's top-right corner. For more information, see the Navigator section in this chapter.
- **Auto Navigator:** The Navigator only appears when one or more nodes is outside the visible area of the Node Editor. For more information, see the Navigator section in this chapter.
- **Build Flow Vertically/Horizontally:** Node trees can be built either horizontally from left to right or vertically from top to bottom. Enabling one of these options determines whether new nodes are added beneath the current node or to the right of the current tool.
- **Orthogonal/Direct Pipes:** Use these two options to decide whether connections between nodes are drawn as Direct (straight) lines or Orthogonal (bent) lines.

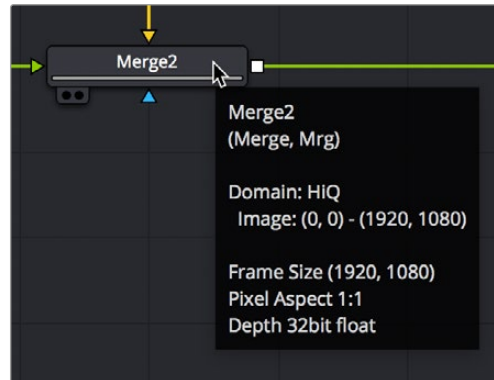
Node Tooltips and the Status Bar

Even in simple node trees, it's easy to forget some essential detail about the nodes in your comp. To help you figure out what everything's for, you can hover the pointer over any node in the Node Editor to display information in the Status bar at the bottom of the Node Editor consisting of that node's name, frame size, pixel aspect, resolution, and color depth.

Merge2 - Frame Size (1920, 1080), Pixel Aspect 1:1, Depth 32bit float

The Status bar located beneath the Node Editor.

If you wait a few moments later, a more elaborate presentation of the same information appears within a floating tooltip in the Inspector. This tooltip gives you additional information about the Domain (Image and DoD) and the data range used by that clip.



The floating tooltip showing node information that appears within the Node Editor.